

APT_{BENCH}: BENCHMARKING CROSS-PATIENT CELL TYPING FROM SINGLE-CELL APTAMER PROFILES

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Paper under double-blind review

ABSTRACT

Which cell identities can aptamer profiles recover in new patients, and what do they add beyond RNA? **APT_{BENCH}** specifies this comparison using 361,792 paired cells from 40 participants, 293 aptamer measurements, 5 lineages, 27 RNA-derived subtypes, and fixed patient-disjoint folds. Two findings emerge. First, a separately evaluated full-training-budget MLP achieves fine subject-balanced Macro-F1 of 0.141, rising to 0.385 with true-lineage decoding, compared with 0.180 for a training-only conditional-proportion prior. This comparison shows additional discrimination relative to the tested training-only lineage priors; matched-budget APT/RNA comparisons reveal substantial lineage heterogeneity. Second, adding paired APT improves MLP fine Macro-F1 by 0.0180 and 0.0165 over RNA references using 2,000 and 5,000 training-selected genes, respectively. Yet the overall advantage over APT shuffled within patient and true lineage remains statistically unresolved. Thus an increment over these RNA references is established without establishing finer-grained pairing specificity. Classical and neural APT-only baselines anchor a reusable new-patient comparison, separating recoverable identity from claims of an intrinsic assay limit or a biological mechanism.

1 INTRODUCTION

Beyond transcriptomic profiling, single-cell technologies increasingly measure complementary molecular layers within the same cell, including cell-surface features that are not fully captured by RNA abundance (Stoeckius et al., 2017; Hao et al., 2021). Aptamers offer one route to these measurements: short oligonucleotide probes selected for target binding, whose sequences enable aptamer-binding profiles to be read out alongside the transcriptome of the same cell. Apt-seq demonstrated joint aptamer and RNA profiling and discrimination of cell populations in defined cell-line mixtures (Delley et al., 2018). Subsequent work broadened these capabilities: Aptomics enabled joint profiling of surface proteins, glycans, and RNA in clinical specimens (Wu et al., 2025), while SPARK-seq integrated CRISPR perturbations with aptamer and RNA sequencing to support aptamer target identification and binding-kinetic characterization (Luo et al., 2026). These advances establish aptamer profiling as a tool for characterizing cellular heterogeneity and motivate its evaluation as a predictive input for cell identity. A central question is how reliably aptamer-binding (APT) features alone can predict annotated cell identities in patients who contributed no data to model development. Evaluating this capability can guide the development of APT-based annotation models and clarify which cell-identity distinctions a given aptamer panel supports in new patients.

Demonstrating cell-population discrimination does not by itself establish how a fixed aptamer panel will perform in patients excluded from model development. RNA-based annotation studies have adopted donor-held-out evaluation to assess generalization (Fischer et al., 2024), while recent benchmarking efforts provide standardized single-cell tasks and systematic comparisons of multimodal methods, including cell-type classification (Luecken et al., 2025; Liu et al., 2025). These methodological foundations motivate a focused evaluation of APT through two complementary questions. First, which annotated cell lineages and subtypes can APT profiles alone predict in new patients? Second, when RNA is available, does adding paired APT improve prediction over a specified RNA-only reference? These questions assess APT’s standalone performance and its incremental predictive value under a defined learning setup, respectively. Paired APT and RNA measurements allow both questions to be evaluated using the same cells, reference annotations, and patient partitions. A com-

mon evaluation setting is therefore needed to characterize both forms of predictive utility under the same cross-patient prediction target.

Answering these questions requires an evaluation that distinguishes generalization, hierarchical prediction errors, and the sources of multimodal gains. First, cells from the same patient are not independent biological replicates (Zimmerman et al., 2021). Evaluating prediction in new individuals therefore requires patient separation across training, model selection, and testing (Fischer et al., 2024). Second, cell-type annotations are hierarchically organized (Cultrera di Montesano et al., 2026). An overall subtype score can obscure whether errors arise across broad lineage boundaries or among subtypes within the same lineage. Providing the true lineage for diagnostic evaluation narrows the candidate set, so any resulting improvement should be interpreted relative to a baseline given the same lineage information. Third, improved prediction with RNA+APT does not by itself establish that the gain depends on the original pairing between individual cells’ RNA and APT profiles. Patient- or lineage-associated structure may also contribute, motivating comparisons that disrupt individual cell pairings while preserving these broader groupings. Such controls can help assess whether the original pairing provides additional predictive value beyond the retained structure. Together, these considerations motivate patient-disjoint evaluation, hierarchy-aware error analysis, and controlled comparisons that clarify the role of cell pairing in multimodal prediction.

To address these challenges, we introduce **APTBench**, a patient-disjoint benchmark for cell-identity prediction using 361,792 paired RNA–APT profiles from 40 participants and 293 APT features. RNA-derived annotations define a common reference of 5 lineages and 27 subtypes. 5 fixed patient folds support evaluation in held-out individuals, with subject-balanced pooled Macro-F1 assigning equal total weight to each patient in the pooled confusion matrix. Classical and neural APT-only models establish reference performance on these folds. To characterize hierarchical errors, we analyze cross-lineage and within-lineage confusions and perform diagnostic decoding constrained by the true lineage. We compare this diagnostic performance with conditional-prior baselines estimated only from training data and given the same lineage information. In a separate multimodal evaluation with matched training budgets across input conditions, we compare RNA-only and RNA+APT models and assess the role of RNA–APT pairing by shuffling APT profiles within patients or within patient–lineage groups. Together, these analyses characterize APT’s standalone predictive performance, the structure of its errors, and its incremental value relative to RNA under a shared cross-patient prediction target. Our work makes the following contributions:

- We establish a cross-patient APT cell-typing benchmark with specified inputs, annotations, patient splits, metrics, and baseline results to support subsequent method comparison.
- We characterize hierarchical prediction failures. Under the evaluated MLP recipes, true-lineage decoding yields higher overall subtype SB-Macro-F1 than the tested training-only conditional priors. Lineage-specific comparisons against corresponding priors and RNA references reveal variation obscured by the overall score.
- We demonstrate predictive gains from adding APT to the tested RNA-only references. In the 2,000-gene setting, the overall advantage of RNA+paired APT over RNA+APT with APT shuffled within patient–true-lineage groups remains statistically unresolved.

2 RELATED WORK

Single-cell aptamer profiling. Single-cell aptamer studies investigate cell discrimination, biological characterization, and probe discovery (Xu et al., 2024; Wu et al., 2024). Apt-seq demonstrated paired aptamer and transcriptome measurements and cell discrimination from binding profiles (Dellley et al., 2018). Aptomics extended joint profiling to cell-surface proteins, glycans, and mRNA, examining tumor heterogeneity in clinical samples and glycosylation changes during T-cell differentiation (Wu et al., 2025). SPARK-seq combines CRISPR-based surface-protein perturbations with single-cell mRNA and aptamer sequencing to identify aptamer targets and characterize binding kinetics (Luo et al., 2026). **APTBench** builds on these measurement capabilities and biological applications to evaluate fixed-panel APT-based cell typing.

Cell annotation and hierarchical prediction. CellTypist and SingleR exemplify supervised and reference-based cell annotation (Domínguez Conde et al., 2022; Aran et al., 2019), while scGPT

and Geneformer learn representations from large transcriptomic corpora for downstream fine-tuning (Cui et al., 2024; Theodoris et al., 2023). CHETAH and scHPL organize reference populations into classification hierarchies to accommodate different annotation resolutions (De Kanter et al., 2019; Michielsen et al., 2021). Hierarchical cross-entropy (HCE) incorporates ancestor–descendant label relationships into the training objective (Cultrera di Montesano et al., 2026). **APT_{Bench}** evaluates classical and hierarchy-aware reference models; its true-lineage oracle and conditional priors serve as separate diagnostics with access to the same reference lineage.

Multimodal integration and predictive utility. WNN learns cell-specific modality weights for integrated neighborhood analysis (Hao et al., 2021), while MOFA+, totalVI, and MultiVI provide latent-variable frameworks for joint representation learning and downstream analysis (Argelaguet et al., 2020; Gayoso et al., 2021; Ashuach et al., 2023). Liu et al. (2025) benchmarked integration methods across modality and batch configurations on tasks including classification, clustering, batch correction, and imputation. **APT_{Bench}** examines APT’s incremental predictive utility by comparing RNA+APT with specified RNA-only references under controlled training settings. Shuffling APT profiles within patients or patient–reference-lineage groups further assesses whether predictive gains depend on the original cell-level pairing beyond the preserved group structure.

Single-cell benchmarking and generalization. Annotation and integration benchmarks provide reusable workflows for systematic method comparison (Abdelaal et al., 2019; Luecken et al., 2022), and Open Problems organizes evaluation around defined tasks, datasets, method interfaces, and metrics (Luecken et al., 2025). Within-individual cell dependence is documented in studies of pseudoreplication (Zimmerman et al., 2021), and scTab provides a precedent for donor-wise annotation evaluation (Fischer et al., 2024). Targeting cell-identity prediction in new patients, **APT_{Bench}** specifies a fixed APT panel, paired RNA–APT measurements, an RNA-derived reference hierarchy, fixed patient-disjoint splits, and patient-balanced scoring. This setting establishes common reference results for APT-based cell typing, complemented by diagnostics of hierarchical errors and modality pairing.

3 BENCHMARK SETTING

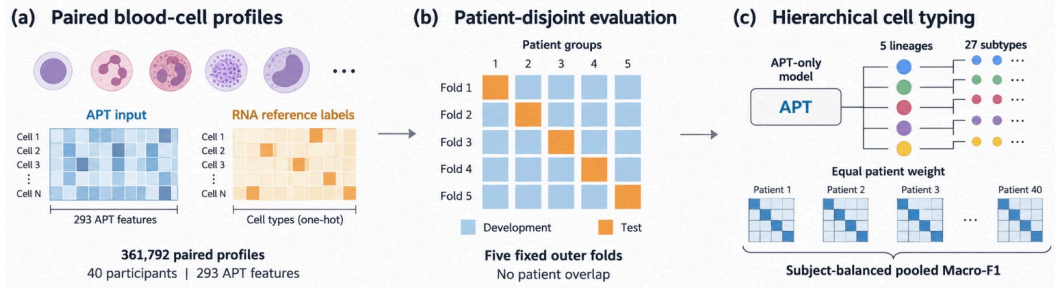


Figure 1: The primary **APT_{Bench}** contract. Patient-disjoint outer folds evaluate cell typing in new individuals; subject-normalized confusion pooling assigns equal total scoring weight to each patient.

3.1 BLOOD PROFILES AND REFERENCE LABELS

APT_{Bench} is derived from peripheral-blood specimens from 40 participants in five cancer groups (CRC, GC, HCC, PC, and BTC) and healthy controls, with 6–8 participants per group. The paired assay measures transcriptomic state and 293 aptamer features on the same cells. Transcriptomic counts were processed with Seurat (Hao et al., 2021); predicted doublets and low-quality cells were removed, and clusters were manually annotated using canonical markers from prior PBMC and pan-cancer studies (McGinnis et al., 2019; Zhang et al., 2024; Yang et al., 2024). These annotations define 5 lineages and 27 subtypes. They are operational reference labels, not independently measured biological truth.

The linked post-QC cohort contains 374,854 cells. Excluding the uncertain *Unknown* subtype leaves 361,792 benchmark cells. Each subtype has one fixed parent lineage. Disease is a six-class patient

attribute, not the root of the cell hierarchy: any given lineage can occur in multiple disease groups. We evaluate the lineage-to-subtype hierarchy without imposing a disease-to-lineage-to-subtype label tree. The construction checklist and unresolved acquisition and annotation fields are recorded in Appendix B.1.

3.2 PATIENT-DISJOINT CELL-TYPING CONTRACT

The primary task is cell typing in new patients. **Cell identity** predicts lineage and subtype using APT alone at inference. All unified main-table recipes use cell-identity supervision only. Disease-based contrastive supervision is evaluated separately for Flat and Soft-cascade; it is not an input required at inference. Open-set subtype discovery is outside the contract; all primary cell tasks use the fixed five-lineage/27-subtype ontology.

5 frozen disease-stratified folds each hold out eight patients. All cells from a patient remain together. Benchmark-fitted transforms, model selection, and fitting use development patients. The unified main comparison uses raw APT counts followed by log1p and fit-partition standardization. The separately retained historical provided-expression diagnostics have an upstream transformation provenance limitation (Appendix B.10). We pool predictions across the five out-of-fold (OOF) partitions. The release interface records cell and patient IDs, ontology, folds, predictions, and metrics (Appendix B.2). Full data access and a portable end-to-end evaluator remain release requirements.

3.3 METRICS AND STATISTICAL UNITS

Coarse and fine subject-balanced pooled Macro-F1 (SB-Macro-F1) are the primary cell metrics. For patient p , let $C^{(p)}$ be the confusion matrix over a fixed K -class ontology and n_p its number of evaluated cells. Define

$$\bar{C} = \frac{1}{P} \sum_{p=1}^P \frac{C^{(p)}}{n_p}, \quad \text{SB-MacroF1} = \frac{1}{K} \sum_{k=1}^K \frac{2\bar{C}_{kk}}{\sum_j \bar{C}_{kj} + \sum_i \bar{C}_{ik}}. \quad (1)$$

A zero denominator contributes zero. Each patient’s entire matrix, not each class row, is normalized to unit mass. This differs from cell-weighted pooled Macro-F1 and from mean within-subject Macro-F1. Unified models use this scoring convention for validation and final reporting. Historical frozen references are retrospectively re-scored without checkpoint reselection.

Main-table intervals use 5,000 resamples and explicitly label their target: P denotes patient uncertainty for deterministic LR, and SP additionally resamples the three training seeds of each randomized model. Each SP replicate averages three seed-specific scores, using the same patient draw across them. Both are pointwise and conditional on the fixed fold design; neither repeats model selection. The comparison is descriptive, and an individual-score interval is not a pairwise superiority test. Full resampling order is given in Appendix B.9. Historical and paired-input diagnostic intervals retain their separately specified resampling protocols.

The paired-RNA probe also uses subject-balanced pooled Macro-F1, but has a different training budget and preprocessing protocol; its scores must not be interpreted as additional architecture rows in the main table. The experimental protocol is in Appendix B.3; scoring-unit comparisons are in Appendix B.7.

4 BENCHMARK REFERENCE MODELS

APT-only comparisons. LR and XGBoost provide linear and nonlinear tabular references. Plain MLP, Flat and HCE Transformers, and Soft-cascade provide neural references. The unified v2 main comparison uses raw-count log1p inputs, training-only standardization, the same 24/8 inner training/validation patients, and fresh refits on all 32 development patients. All available annotated cells are used. Appendix B.9 specifies model-specific search grids, exact fold counts, three randomized refit seeds and realized budgets. LR is deterministic; LR/XGBoost select coarse and fine tasks independently, whereas neural coarse outputs accompany fine-selected fits. Common data and selection rules do not imply identical optimization or compute costs. Historical Linear SVM, Random Forest and reference-correlation results remain appendix references, not additional rows of this unified comparison.

Hierarchy-aware reference. Soft-cascade embeds aptamer identities and values and combines a global subtype head with lineage-specific experts weighted by coarse probabilities. Eq. 6 specifies the routing form; v2 dimensions and objectives are given in Appendix B.9. Ordinary fine predictions are not restricted to the predicted parent. Flat bypasses the cascade; HCE uses a two-level hierarchical loss adaptation (Cultrera di Montesano et al., 2026), with parent probabilities decoded from subtree mass. No main-table model uses disease supervision. A separate 2×2 comparison adds the same cross-disease contrastive term to Flat and Cascade. It tests the whole cascade structure and added supervision under a shared search protocol, not individual routing components. The earlier architecture, loss and component matrix in Appendix A describe historical references and should not be substituted for v2 settings.

Input compatibility. Gene-based pretrained models such as scGPT and Geneformer (Cui et al., 2024; Theodoris et al., 2023) do not directly accept anonymous aptamer feature IDs without a verified mapping or a separately trained adapter. Supplying paired RNA would change the input contract. This distinguishes an undefined direct APT input from an adaptable but unevaluated model; it does not justify claiming superiority over transcriptomic annotation methods.

Matched-input diagnostics. A separate matched-input probe varies APT and RNA while holding its training budget and patient folds fixed. Training-only subtype priors and true-lineage masking provide candidate-set controls. APT shuffling within patient, or within patient and true lineage, is a privileged diagnostic rather than a deployable model input (Appendix B.3). RNA also supplies the reference annotations, so this is an internal reconstruction comparison, not independent label validation. HVG selection and scaling are fitted separately within each inner and outer training partition. Table 5 gives input transforms, training weights, optimization settings and selection grids. The existing provided-expression MLP oracle/prior and failure analyses are also retained as a separate diagnostic recipe; they do not use the v2 MLP fits.

5 RESULTS

5.1 REFERENCE PERFORMANCE IN NEW PATIENTS

Table 1 reports APT-only typing on the same 40 held-out patients. Under the unified v2 protocol, fine SB-Macro-F1 is 0.134 for LR and XGBoost, 0.137 for MLP, 0.147 for Flat, 0.146 for HCE and 0.148 for Cascade without disease supervision. The prespecified descriptive Flat-minus-MLP difference is $+0.0100$ (paired 95% CI $[-0.0047, 0.0234]$), and Cascade-minus-Flat is $+0.0011$ ($[-0.0093, 0.0093]$). These intervals do not establish superiority or equivalence. Models share input transforms, inner partitions and fresh outer refits, but retain model-specific objectives and optimization budgets (Appendix B.9).

Model	CI	Fine SB-F1 $\uparrow$	Coarse SB-F1 $\uparrow$
Logistic Regression	P	0.134 [0.114, 0.146]	0.296 [0.276, 0.315]
XGBoost	SP	0.134 [0.118, 0.148]	0.326 [0.304, 0.348]
Plain MLP	SP	0.137 [0.121, 0.152]	0.333 [0.307, 0.356]
Flat Transformer	SP	0.147 [0.127, 0.162]	0.342 [0.317, 0.365]
HCE Transformer	SP	0.146 [0.126, 0.160]	0.339 [0.316, 0.362]
Soft-cascade (no disease)	SP	0.148 [0.129, 0.161]	0.336 [0.311, 0.359]

Table 1: Unified v2 APT-only reference comparison on the same 361,792 OOF cells and 40 patients. All models use raw-count $\log_{1p}$ inputs, training-only standardization, the same inner-patient split, and fresh refits on all 32 development patients. No main-table model uses disease supervision. Values are means across three training seeds (LR: one deterministic fit) with 5,000-resample pointwise 95% intervals: P resamples patients; SP jointly resamples patients and seeds. LR/XGBoost select the two tasks separately; neural coarse scores are auxiliary outputs of fine-selected fits. HCE coarse scores use subtree probability mass. Model-specific search spaces and the selection-conditioned uncertainty target are given in Appendix B.9.

Recipe	Disease	Fine SB-F1	Paired difference [95% CI]
A: Flat	No	0.147 [0.127, 0.162]	—
B: Flat	Yes	0.148 [0.126, 0.163]	B−A: 0.0006 [-0.0057, 0.0071]
C: Cascade	No	0.148 [0.129, 0.161]	C−A: 0.0011 [-0.0093, 0.0093]
D: Cascade	Yes	0.150 [0.128, 0.166]	D−C: 0.0017 [-0.0087, 0.0123]

Table 2: Unified v2 2×2 comparison of the whole cascade structure and cross-disease contrastive supervision. A and C reuse the main-table fits; B and D add the same contrastive term. Shared encoder, inputs and search rules are held fixed, but each recipe selects its own validation optimum. C−A is not a routing-only effect. The interaction (D−C)−(B−A) is 0.0012 [-0.0087, 0.0110]. Intervals are paired, pointwise and exploratory, not multiplicity-adjusted tests.

Structure and extra supervision. Table 2 separates the full cascade structure from cross-disease contrastive supervision. Adding contrastive supervision changes fine SB-F1 by +0.0006 for Flat and +0.0017 for Cascade; their paired intervals and the interaction interval include zero. This does not establish that these recipes are equivalent or that hierarchy is unhelpful. In particular, Cascade-minus-Flat changes the global/expert/routing combination, not soft routing alone.

Can predicted parents realize the oracle gain? Frozen-score diagnostics in Table 13 compare ordinary subtype prediction with predicted-parent masking (D1), probability-consistent combination (D2), and true-parent masking (D4). For the v2 MLP, fine SB-F1 is 0.137, 0.131, 0.134 and 0.376 respectively. Thus these two deployable readout rules do not realize the oracle gain for this recipe. This is not a test of all possible trained hierarchical classifiers. The prior-controlled analyses below use their original, separately identified fits.

Five-lineage and 27-subtype tasks differ in cardinality, prevalence and label boundaries. Their raw F1 difference alone does not identify an APT resolution limit. The following prior-controlled comparisons instead ask what remains recoverable within a known lineage. Full subtype reporting and the supporting split-protocol comparison are retained in Appendices B.6 and B.7.

5.2 WITHIN-LINEAGE IDENTITY IS RECOVERABLE, BUT HETEROGENEOUS

True-lineage decoding masks each frozen fine score vector to the correct parent’s children. Both the model and the training-only conditional priors receive this privileged label. For the earlier full-budget provided-expression MLP (Table 15, not the raw-count v2 MLP), ordinary fine F1 is 0.141 and oracle F1 is 0.385, versus 0.148 for conditional majority decoding and 0.180 for the conditional-proportion expected-confusion baseline. The oracle-minus-expected-prior difference is +0.205 (joint 95% CI [0.174, 0.217]). Thus APT supports discrimination beyond these candidate-set controls under this recipe; the comparison does not measure all information in the assay.

Saved-score diagnostics further locate the errors: in this full-budget MLP, cross-parent and within-parent subtype errors occupy 49.4% and 17.3% of patient-balanced cell mass, respectively. Oracle corrects 63.2% of cross-parent errors when correction fractions are first computed within each patient. The two largest remaining directed confusions are CD8 Tem-1 to CD4 + Tcm-1 and the reverse, ranked first and second in all three seeds. Both subtypes occur in all 40 patients. These error rates complement, rather than additively decompose, Macro-F1; complete diagnostics appear in Appendix B.11.

The matched-input probe holds budget and recipe fixed across modalities (Figure 2). MLP ordinary/oracle APT scores are 0.093/0.309, RNA oracle is 0.672, and the expected prior is 0.182. The APT oracle-minus-prior difference is +0.127 ([0.101, 0.139]). Relative to each lineage’s own conditional prior and RNA reference, APT reconstruction differs across lineages. Their absolute F1 values are not a common biological-resolution scale: child-class counts and prevalence also differ. For example, APT/RNA oracle scores are 0.849/0.839 for NK but 0.191/0.535 for T cells. Knowledge of the parent therefore does not uniformly resolve the RNA reconstruction gap. All lineages and both model families are shown, without asserting an intrinsic modality ranking.

Budget and recipe matter. The full-budget MLP uses all development cells, a 256/128-unit network and AdamW with selected stopping; the matched probe caps training at 1,000 cells per patient

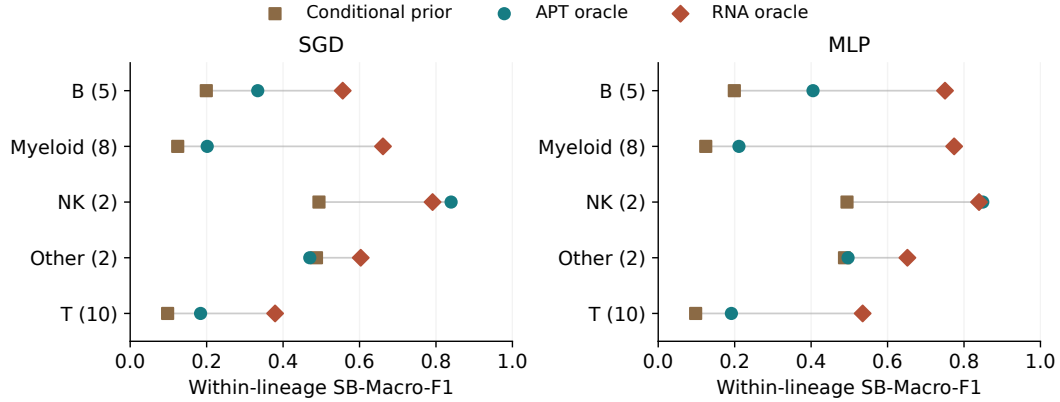


Figure 2: Matched-budget within-lineage reconstruction with 2,000 training-selected RNA HVGs. APT oracle, RNA oracle and conditional priors use the same capped training subsets, patient folds and true-lineage information. Points are means of three seed-specific scores, not F1 of seed-pooled predictions; lines only connect methods within a lineage. Each patient’s matrix is normalized within the displayed lineage before pooling over eligible patients and scoring its fixed children (38 patients for Other; 40 elsewhere). Consequently, these lineage scores cannot be averaged to reconstruct the All row of Table 6 from this same probe. Row labels give the number of child subtypes. These are descriptive point estimates, not superiority tests. Expected-prior F1 is computed from an expected confusion matrix.

and uses a 64-unit MLP for 40 epochs. Other has oracle F1 of 0.886 in the former but 0.497 in the latter (conditional expected priors 0.490/0.487). The recipes also differ in APT preprocessing (Table 5): $\log_{1p}$ of raw counts in the matched probe versus provided APT expression without an added $\log$ transform in the full-budget MLP, each followed by training-set standardization. Their score differences therefore cannot be attributed to training-cell budget alone. Neither result justifies declaring Other universally identifiable or unidentifiable. The full-budget lineage plot and neural-reference replays are supporting evidence in Appendices B.4–B.5.

5.3 PAIRED APT IMPROVES ON RNA, WITHOUT RESOLVED PAIRING SPECIFICITY

Across three training seeds under the matched probe, MLP fine SB-Macro-F1 increases from 0.589 for RNA to 0.607 for RNA+APT: $+0.0180$ (joint 95% CI $[0.009, 0.030]$). At 5,000 training-selected HVGs, it increases from 0.596 to 0.612: $+0.0165$ ($[0.009, 0.029]$). SGD also improves at both widths (Table 3). The increment is therefore not specific to the 2,000-HVG setting, but remains relative to the tested RNA representations and models. Both widths use the capped-budget probe, not a full-cell-budget RNA model; increasing HVGs does not control network capacity or training sufficiency.

RNA HVGs	Comparator to paired RNA+APT	SGD Δ [95% CI]	MLP Δ [95% CI]
2,000	RNA	+0.0185 [0.0034, 0.0332]	+0.0180 [0.0092, 0.0297]
2,000	RNA + noise	+0.0394 [0.0240, 0.0537]	+0.0341 [0.0208, 0.0448]
2,000	Patient shuffle	+0.0059 [−0.0049, 0.0165]	+0.0088 [0.0014, 0.0145]
2,000	Patient $\times$ lineage shuffle	+0.0064 [−0.00437, 0.01657]	+0.0064 [−0.00018, 0.01180]
5,000	RNA (width sensitivity)	+0.0217 [0.0104, 0.0339]	+0.0165 [0.0093, 0.0289]

Table 3: Overall fine SB-Macro-F1 differences, positive favoring paired RNA+APT. Shuffle comparators append shuffled APT to RNA. All 2,000-HVG contrasts use the same targeted experiment: three training seeds and three realizations per seed for each shuffle type. The 5,000-HVG row is the separate matched-width RNA comparison, not a conditional-shuffle test. Scores are computed per fit before averaging; 2,000 paired seed/patient bootstrap replicates additionally resample shuffle realizations where applicable. Intervals are pointwise and exploratory. Both patient-by-lineage-shuffle intervals include zero. Differences are calculated before rounding; displayed marginal scores need not subtract to the displayed difference.

What the controls establish. In the targeted 2,000-HVG experiment, MLP exceeds RNA+noise by 0.0341 and patient-shuffled APT by 0.0088 (joint 95% CI [0.0014, 0.0145]). This interval re-samples patients, training seeds and shuffle realizations; it is not a patient-only interval conditional on one fitted model. The latter comparison uses three shuffle draws per training seed, as does the more restrictive patient-by-lineage control. Paired minus patient-by-lineage-shuffled APT is +0.0064 ([−0.00018, 0.01180]); SGD’s corresponding interval also includes zero. This is an unresolved overall contrast, not equivalence or evidence of no pairing information. All control contrasts in the table come from the same targeted experiment; the separate one-shuffle-per-seed width grid is labeled as a sensitivity analysis in the appendix. Lineage-specific contrasts remain exploratory supporting analyses (Appendix B.3).

6 DISCUSSION

APT_{Bench} separates two empirical questions: the recoverability of cell identity at different hierarchy levels, and the incremental contribution of paired APT beyond a specified RNA reference. The completed MLP oracle and training-prior comparison shows additional discrimination relative to the tested training-only lineage priors. Its heterogeneity across lineages cautions against summarizing APT as uniformly informative or uninformative. The paired-input increment asks a different question. It persists over the two tested RNA widths, but the repeated patient-by-lineage-shuffle contrast remains unresolved overall. Improvements over RNA alone therefore do not establish finer-grained pairing specificity. Patient-disjoint testing and subject-balanced scoring provide the common evaluation foundation, not an additional claim of methodological novelty.

A reusable comparison setting. The resource specifies an APT input, operational ontology, fixed patient folds, metrics, and reference results. Soft-cascade is a hierarchy-aware reference, not a claimed algorithmic advance. These references establish comparison points for improving APT cell typing without changing the input modality, ontology, or held-out patient partitions. Complete data access, permissions, annotation provenance, and a portable evaluator remain necessary release requirements. The unified comparison supplies full-cell-budget classical and neural references, with separate structure/supervision contrasts. Neither the Flat-minus-MLP nor Cascade-minus-Flat interval establishes superiority or equivalence. More comprehensive APT-compatible annotation base-lines remain useful extensions.

Boundaries of the evidence. The APT cohort contains 40 participants and RNA-derived reference labels. Independent annotation checks and additional APT cohorts are needed to assess broader generalization. Patient-disjoint testing does not remove disease-correlated acquisition effects; this

manuscript makes no disease classification or clinical-validation claim. RNA supplies the reference labels, and its apparent difficulty depends on features, preprocessing, and the fitted reference. An APT increment over either 2,000 or 5,000 HVGs may reflect restricted RNA representation; it is not evidence of irreducible information absent from RNA. The capped probe does not establish an increment over a full-cell-budget, capacity-controlled RNA reference. Likewise, oracle-versus-prior performance does not substitute for an APT-only, patient-by-lineage-shuffled control, or guarantee that predicted-lineage decoding realizes the oracle gain. The v2 frozen-score tests address two such deployable rules, without covering all possible trained hierarchical models; an APT-only conditional shuffle remains a distinct unanswered comparison. True-lineage masking and shuffling are privileged diagnostics, and the latter does not identify a causal or molecular mechanism. Three training seeds and pointwise resampling intervals also provide limited uncertainty evidence. The outer folds have been examined repeatedly during this study; fixed follow-up analyses on them are not untouched confirmatory tests. Their overlapping development sets also limit interpreting OOF bootstrap intervals as uncertainty for a learning algorithm on an independently sampled cohort.

7 CONCLUSION

APTBench establishes a patient-disjoint evaluation setting for cell typing from single-cell aptamer profiles, with a fixed label ontology, explicit subject-balanced scoring, and classical and neural reference results. It asks which hierarchy levels are recoverable in new patients and whether APT contributes additional typing information beyond an RNA reference. Training-prior-controlled oracle decoding reveals usable but heterogeneous within-lineage discrimination under the tested model. The repeated paired probe improves over the tested RNA-only references at both widths, while its overall advantage over patient-by-lineage-shuffled APT remains unresolved. Distinguishing these findings avoids equating improved reference reconstruction with established subtype-specific complementarity.

AI USE STATEMENT

Generative-AI assistance through ChatGPT and Codex was used in the revision workflow for research-design suggestions, literature search, manuscript editing, analysis-code development, debugging, experiment orchestration, and scientific figure preparation. AI-generated suggestions were not themselves experimental evidence; reported computational results are tied to saved run artifacts. **TODO: Authors: confirm the complete tool/model inventory and versions, describe the human checks actually performed on code, citations, figures, and statistical claims, and affirm responsibility for the final submission. Automated checks do not substitute for this author verification.**

ETHICS STATEMENT

APTBench is intended for research evaluation, not clinical decision-making. The small single-cohort sample and unresolved acquisition effects preclude claims of clinical validity or demographic fairness. Strong patient-specific signals also create a privacy concern: removing direct identifiers alone does not establish that cell profiles cannot be linked to their source individuals. Any data release must respect participant consent and applicable access terms. **TODO: Data owners: supply the actual ethics approval or exemption and consent terms; confirm de-identification procedures, permitted sharing/access controls, and conflicts of interest. These facts have not been verified from the available analysis artifacts; preserve reviewer anonymity when providing documentation.**

REPRODUCIBILITY STATEMENT

The repository records the fixed five-fold partition in `benchmark/splits/patient_folds.json`, selected artifact checksums and structural audits in `benchmark/release_audit/`, and artifact-level reproduction commands in `REPRODUCIBILITY.md`. The latter distinguishes rebuilding figures/PDFs from rerunning models. Per-experiment protocol files record sampling settings and seeds. The findings-focused display builder reads completed aggregate results without retraining; its source-to-table mapping and the exact bootstrap audit are recorded in `analysis/FINDINGS_FOCUS_REVISION.md`. A snapshot of the local analysis environment

is provided, but is not a lockfile for all historical training runs. The unified reference tables are generated by `analysis/unified_v2/build_paper_tables.py` from validated result summaries; the same directory records the frozen protocol, stage inventory and HCE readout correction without retraining. **TODO: Finalize anonymous reviewer access, data permissions and licenses, archival versioning, original training environments, and a portable raw-data-to-results evaluator. The present repository is not yet a complete independently reproducible benchmark release.**

AUTHOR CONTRIBUTIONS

TODO: Add author contributions using CRediT roles (for example: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and funding acquisition). Preserve anonymity in the review version if required.

ACKNOWLEDGMENTS

TODO: Add funding sources and grant numbers, computational resources, data or clinical contributors, and non-author assistance; verify whether any acknowledgment must be omitted or anonymized during double-blind review.

REFERENCES

- Tamim Abdelaal, Lieke Michielsen, Davy Cats, Dylan Hoogduin, Hailiang Mei, Marcel JT Reinders, and Ahmed Mahfouz. A comparison of automatic cell identification methods for single-cell rna sequencing data. *Genome biology*, 20(1):194, 2019.
- Dvir Aran, Agnieszka P Looney, Leqian Liu, Esther Wu, Valerie Fong, Austin Hsu, Suzanna Chak, Ram P Naikawadi, Paul J Wolters, Adam R Abate, et al. Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage. *Nature immunology*, 20(2):163–172, 2019.
- Ricard Argelaguet, Damien Arnol, Danila Bredikhin, Yonatan Deloro, Britta Velten, John C Marioni, and Oliver Stegle. Mofa+: a statistical framework for comprehensive integration of multi-modal single-cell data. *Genome biology*, 21(1):111, 2020.
- Tal Ashuach, Mariano I Gabitto, Rohan V Koodli, Giuseppe-Antonio Saldi, Michael I Jordan, and Nir Yosef. Multivi: deep generative model for the integration of multimodal data. *Nature methods*, 20(8):1222–1231, 2023.
- Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scgpt: toward building a foundation model for single-cell multi-omics using generative ai. *Nature methods*, 21(8):1470–1480, 2024.
- Sebastiano Cultrera di Montesano, Davide D’Ascenzo, Srivatsan Raghavan, Ava P Amini, Peter S Winter, and Lorin Crawford. Improving atlas-scale single-cell annotation models with hierarchical cross-entropy loss. *Nature Computational Science*, 6(3):243–249, 2026.
- Jurrian K De Kanter, Philip Lijnzaad, Tito Candelli, Thanasis Margaritis, and Frank CP Holstege. Chetah: a selective, hierarchical cell type identification method for single-cell rna sequencing. *Nucleic acids research*, 47(16):e95–e95, 2019.
- Cyrille L Delley, Leqian Liu, Maen F Sarhan, and Adam R Abate. Combined aptamer and transcriptome sequencing of single cells. *Scientific reports*, 8(1):2919, 2018.
- César Domínguez Conde, Chao Xu, Louie B Jarvis, Daniel B Rainbow, Sara B Wells, Tamir Gomes, Sarah K Howlett, Ondrej Suchanek, Krzysztof Polanski, Hamish W King, et al. Cross-tissue immune cell analysis reveals tissue-specific features in humans. *Science*, 376(6594):eab15197, 2022.
- Felix Fischer, David S Fischer, Roman Mukhin, Andrey Isaev, Evan Biederstedt, Alexandra-Chloé Villani, and Fabian J Theis. sctab: Scaling cross-tissue single-cell annotation models. *Nature Communications*, 15(1):6611, 2024.

- Adam Gayoso, Zoë Steier, Romain Lopez, Jeffrey Regier, Kristopher L Nazon, Aaron Streets, and Nir Yosef. Joint probabilistic modeling of single-cell multi-omic data with totalvi. *Nature methods*, 18(3):272–282, 2021.
- Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. In *Advances in Neural Information Processing Systems*, volume 34, pp. 18932–18943, 2021.
- Yuhan Hao, Stephanie Hao, Erica Andersen-Nissen, William M Mauck, Shiwei Zheng, Andrew Butler, Maddie J Lee, Aaron J Wilk, Charlotte Darby, Michael Zager, et al. Integrated analysis of multimodal single-cell data. *Cell*, 184(13):3573–3587, 2021.
- Chunlei Liu, Sichang Ding, Hani Jieun Kim, Siyu Long, Di Xiao, Shila Ghazanfar, and Pengyi Yang. Multitask benchmarking of single-cell multimodal omics integration methods. *Nature Methods*, 22(11):2449–2460, 2025.
- Malte D Luecken, Maren Büttner, Kridsakorn Chaichoompu, Anna Danese, Marta Interlandi, Michaela F Müller, Daniel C Strobl, Luke Zappia, Martin Dugas, Maria Colomé-Tatché, et al. Benchmarking atlas-level data integration in single-cell genomics. *Nature methods*, 19(1):41–50, 2022.
- Malte D Luecken, Scott Gigante, Daniel B Burkhardt, Robrecht Cannoodt, Daniel C Strobl, Nikolay S Markov, Luke Zappia, Giovanni Palla, Wesley Lewis, Daniel Dimitrov, et al. Defining and benchmarking open problems in single-cell analysis. *Nature Biotechnology*, 43(7):1035–1040, 2025.
- Guoyan Luo, Jia Song, Yongbo Fu, Yuanjie Jiang, Yuan Gao, Zhixing Zhong, Ling Li, Yong Wei, Hao-Ran Jia, Lu Guo, et al. Spark-seq: a high-throughput platform for aptamer discovery and kinetic profiling. *Science*, 391(6780):eadv6127, 2026.
- Christopher S McGinnis, Lyndsay M Murrow, and Zev J Gartner. Doubletfinder: Doublet detection in single-cell rna sequencing data using artificial nearest neighbors. *Cell Systems*, 8(4):329–337.e4, 2019. doi: 10.1016/j.cels.2019.03.003.
- Lieke Michielsen, Marcel JT Reinders, and Ahmed Mahfouz. Hierarchical progressive learning of cell identities in single-cell data. *Nature Communications*, 12(1):2799, 2021.
- Marlon Stoeckius, Christoph Hafemeister, William Stephenson, Brian Houck-Loomis, Pratip K Chattopadhyay, Harold Swerdlow, Rahul Satija, and Peter Smibert. Simultaneous epitope and transcriptome measurement in single cells. *Nature methods*, 14(9):865–868, 2017.
- Christina V Theodoris, Ling Xiao, Anant Chopra, Mark D Chaffin, Zeina R Al Sayed, Matthew C Hill, Helene Mantineo, Elizabeth M Brydon, Zexian Zeng, X Shirley Liu, et al. Transfer learning enables predictions in network biology. *Nature*, 618(7965):616–624, 2023.
- Xiaoqiu Wu, Yuqing Liu, Dengwei Zhang, Jingjing Yu, Mingxin Zhang, Shuwei Feng, Lifei Zhang, Ting Fu, Yamin Tan, Tao Bing, et al. Efficient strategy to discover dna aptamers against low abundance cell surface proteins in scarce samples. *Journal of the American Chemical Society*, 146(39):26667–26675, 2024.
- Xiaoqiu Wu, Xinrui Lin, Xiangqi Ma, Huidong Huang, Dengwei Zhang, Yuqing Liu, Rui Cheng, Jia Song, Lifei Zhang, Yamin Tan, et al. High-throughput multiplexed quantification of molecules by aptamer sequencing (apt-seq) in single cells. *Journal of the American Chemical Society*, 147(24):20394–20405, 2025.
- Liujun Xu, Yawei Feng, Tong Wang, Shenhuan Li, Kangli Xu, Yue Sun, Yi Luo, Yishan Ye, Yan Miao, Yun Dong, et al. Aptamer-based cell-surface profiling with single-cell resolution enables precise cancer characterization. *ccs Chemistry*, 6(1):196–207, 2024.
- Yu Yang, Xueyan Chen, Jieying Pan, Huiheng Ning, Yaojun Zhang, Yufei Bo, Xianwen Ren, Jiesheng Li, Shishang Qin, Dongfang Wang, et al. Pan-cancer single-cell dissection reveals phenotypically distinct b cell subtypes. *Cell*, 187(17):4790–4811, 2024.

Yuanhang Zhang, Xiaorui Zhou, Yu Zhong, Xi Chen, Zeyu Li, Rui Li, Pengfei Qin, Shanshan Wang, Jianhua Yin, Shang Liu, et al. Pan-cancer scRNA-seq analysis reveals immunological and diagnostic significance of the peripheral blood mononuclear cells. *Human Molecular Genetics*, 33(4):342–354, 2024.

Kip D Zimmerman, Mark A Espeland, and Carl D Langefeld. A practical solution to pseudoreplication bias in single-cell studies. *Nature communications*, 12(1):738, 2021.

A HISTORICAL TRANSFORMER REFERENCE IMPLEMENTATION DETAILS

This appendix gives the historical implementation details for the hierarchy-aware reference ladder: a flat Transformer, its HCE counterpart, and Soft-cascade Transformer. The cascade focuses on the cell-level hierarchy, lineage $\rightarrow$ subtype. The reference recipes differ in auxiliary supervision as well as hierarchy structure; their comparison does not isolate individual components. These settings apply to historical exports and frozen-checkpoint diagnostics, not the unified v2 main table. The v2 configuration and its separate 2×2 comparison are specified in Appendix B.9.

A.1 CELL ENCODER

Each cell is represented by a standardized APT vector $x \in \mathbb{R}^{293}$. We embed feature identity and feature value separately. For aptamer index i , we form a token

$$t_i = \text{Emb}(i) + \text{MLP}_{\text{val}}(x_i), \quad i = 1, \dots, 293. \quad (2)$$

A learnable [CLS] token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final [CLS] representation

$$z = \text{Encode}(x) \in \mathbb{R}^{256} \quad (3)$$

serves as the shared cell representation for downstream tasks.

A.2 SOFT-CASCADE REFERENCE

Let $M = 5$ be the number of coarse lineages and $K = 27$ the number of fine subtypes. Each subtype k belongs to exactly one parent lineage $m(k)$. A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \quad (4)$$

with routing distribution

$$p(m \mid x) = \text{softmax}(c/\tau)_m, \quad (5)$$

where $\tau = 1$.

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, Soft-cascade Transformer uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha p(m(k) \mid x) e_{m(k),k}(z) + \gamma \log(\epsilon + (1 - \epsilon)p(m(k) \mid x)), \quad (6)$$

Here $g(z)$ is a global subtype head, $e_{m,k}(z)$ is the expert associated with lineage m , $\alpha = 1.0$, $\epsilon = 0.1$, and γ is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg \max_k s_k(x), \quad \hat{m} = \arg \max_m c_m. \quad (7)$$

A.3 CROSS-DISEASE CONTRASTIVE REGULARIZATION

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\text{Proj}(z)}{\|\text{Proj}(z)\|_2} \quad (8)$$

be a normalized projection. For anchor i , P_i contains same-subtype, different-disease cells, and N_i contains different-subtype cells. Self-pairs and same-subtype/same-disease pairs are excluded. The implementation uses positive probability mass, rather than the standard average of individual positive log-probabilities:

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{|V|} \sum_{i \in V} \log \frac{\sum_{j \in P_i} \exp(u_i^\top u_j / 0.1)}{\sum_{j \in P_i \cup N_i} w_{ij} \exp(u_i^\top u_j / 0.1)}, \quad (9)$$

where $V = \{i : |P_i| > 0\}$. Anchors without positives are excluded; a batch with no valid anchor returns zero loss. Although the code permits extra weight on same-lineage negatives, the inspected resolved configuration uses $w_{ij} = 1$ for every included pair. The contrastive multiplier is zero through epoch 5 and then $0.2 \min(1, (t - 5)/10)$. This disease-supervised objective is not used by Flat or HCE and is not an independently validated mechanism.

A.4 OPTIMIZATION AND MODEL SCOPE

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}} \mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}} \mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t) \mathcal{L}_{\text{contrast}}, \quad (10)$$

where $\mathcal{L}_{\text{fine}}$ and $\mathcal{L}_{\text{coarse}}$ are cross-entropy losses with $\lambda_{\text{fine}} = 1.0$ and $\lambda_{\text{coarse}} = 0.5$. The contrastive weight increases to 0.2 according to the schedule in §A.3.

We optimize the cascade using AdamW with learning rate 10^{-4} and weight decay 10^{-5} , gradient clipping at norm 1.0, batch size 512, and mixed-precision training. The retained training configuration uses a maximum of 50 epochs and patience 10 on validation fine-level cell-weighted (CW) Macro-F1 over the fixed 27 classes. It uses unweighted mini-batch sampling; no independent benefit or harm of sampling weights is inferred from the reference comparison.

Backbone-matched Transformer references. The flat FT-style Transformer reference (Gorishniy et al., 2021) removes the cascade while retaining the APT tokenizer, Transformer dimensions, optimizer, patience and maximum-epoch budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by Soft-cascade Transformer, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same $M + K$ lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices. For HCE (Cultrera di Montesano et al., 2026), let $R_{ij} = 1$ when node j is node i or its descendant, $p = \text{softmax}(q)$ over all nodes, and $s = Rp$. We minimize $-\lambda_{\text{fine}} \log s_{M+k} - \lambda_{\text{coarse}} \log s_m$ for observed subtype k and lineage m . Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. Coarse HCE decoding maximizes subtree scores s_m , not raw internal-node logits; fine decoding maximizes leaf logits. All three Transformer models are evaluated from the checkpoint selected by held-out validation-patient fine CW Macro-F1, not SB-Macro-F1, without an additional post-selection calibration stage. Table 4 makes the remaining objective differences explicit. In particular, only the soft-cascade reference uses disease labels to construct cross-disease contrastive pairs; Flat and HCE use no disease labels and no contrastive term. Flat/HCE use the sklearn observed-label union for validation Macro-F1; Soft-cascade uses all 27 classes. Shared backbone dimensions, data partitions and maximum-epoch limits therefore do not imply identical selection targets or realized training lengths. This is not an objective-matched component ablation (Table 16). We do not attribute any score difference to soft routing, hierarchical loss, or contrastive regularization in isolation.

Component	Flat	HCE	Soft-cascade
APT tokenizer and Transformer dimensions	Same	Same	Same
Patient folds, optimizer, maximum epochs	Same	Same	Same
Validation fine CW-F1 label set	Observed union	Observed union	All 27 classes
Fine-label supervision	Independent CE	Leaf HCE NLL	CE
Coarse-label supervision	Independent CE	Ancestor HCE NLL	CE
Cross-disease supervised contrastive term	No	No	Yes ($w \leq 0.2$)
Disease labels used by cell model	No	No	Contrastive only
Global subtype path and lineage experts	No	No	Yes
Soft lineage routing	No	No	Yes

Table 4: Components of the three neural references. “Same” denotes an identical specification, not shared fitted weights. The models match the tokenizer, encoder dimensions, data folds, optimizer, maximum epochs and patience, but not the precise validation label-set rule or the complete training objective. In particular, disease labels enter only the soft-cascade contrastive pairing rule. Consequently, comparisons among these rows evaluate complete recipes and are not component ablations.

The main cell-identity results are reported in Table 1; the following sections supply their implementation and diagnostic details.

B DATASET AND EVALUATION DETAILS

B.1 DATASET CONSTRUCTION AND DOCUMENTATION CHECKLIST

The available study workflow establishes the high-level sequence used to construct [APT Bench](#): clinical peripheral-blood specimens from five cancer groups and healthy controls are converted to single-cell suspensions; cell-profile and viability quality control precedes a paired single-cell transcriptome and 293-aptamer assay; separate transcriptome and aptamer-barcode libraries are linked at the cell level; and transcriptome-derived clusters and annotations provide the cell labels. For a benchmark release, this overview must be accompanied by the following provenance and protocol fields.

Outstanding disease-confounding validation. **TODO:** Obtain a sample-linked acquisition manifest from the data owner: patient and sample IDs, collection and processing dates, APT assay batch, plate/pool, reagent lot, library-preparation batch, sequencing run, site, and available fresh/frozen and handling records. Audit disease-by-batch overlap before choosing a validation design. Where disease groups have adequate cross-batch support, evaluate patient-disjoint, held-out-batch generalization with all preprocessing and selection restricted to training data. If disease and batch are inseparable, obtain crossed-batch samples or an independent APT cohort; statistical adjustment alone cannot identify their separate contributions. The NC matrices must not be treated as external validation if they are the healthy subset of this cohort. These validations are pending, not completed. Expression and QC matrices alone do not establish acquisition provenance; archived disease analyses cannot establish disease-specific biological effects.

Transcriptomic preprocessing and cell quality control. The transcriptomic count matrix was processed with Seurat following its standard workflow of normalization, feature selection, dimensionality reduction, neighborhood-graph construction, and clustering (Hao et al., 2021). DoubletFinder was used to identify and remove predicted doublets (McGinnis et al., 2019). Low-quality cells were additionally filtered using per-cell UMI count, detected-gene count, and mitochondrial-transcript fraction. These steps preceded cluster annotation and construction of the linked APT benchmark cohort. **TODO:** Report the Seurat and DoubletFinder versions, every parameter and random seed, the exact UMI/gene/mitochondrial thresholds, and cell counts before and after each filter.

Manual annotation and reference-label reliability. After Seurat clustering, investigators manually assigned lineage and subtype names by inspecting canonical marker-gene expression reported in prior PBMC and pan-cancer studies (Zhang et al., 2024; Yang et al., 2024). No quantitative annotation confidence, inter-annotator agreement, or external reference-transfer score is available.

Accordingly, these labels are operational reference annotations, not error-free biological ground truth, and benchmark scores are conditional on this manual taxonomy. The *Unknown* label was assigned when a cluster did not show a coherent canonical marker pattern for the expected PBMC subtypes; this was a manual reject decision rather than a calibrated confidence threshold. Unknown cells are excluded from the primary 27-subtype benchmark. **TODO: Provide the full subtype-by-marker citation table, the number and expertise of annotators, whether they were blinded to disease, the adjudication procedure, and the explicit rule used to assign Unknown.**

Cross-disease semantics and hierarchy mapping. The same 27 subtype labels and definitions were applied across all six disease groups; no subtype was renamed or redefined for a particular disease. This gives the labels a consistent operational meaning, but it does not prove that marker expression or biological state is invariant across diseases. Each subtype is assigned to one of five parent lineages by a manually curated lookup. The exported lookup and structural audit are available under `benchmark/release_audit/`: the existing subtype summary has 27 unique subtypes, five parents, and 361,792 cells, and the fold manifest has five disjoint groups of eight patients. These checks establish internal bookkeeping consistency, not independent biological validity. **TODO: Document independent validation of the subtype-to-lineage mapping, including reviewers, disagreements, adjudication, and any marker-based checks.**

- **TODO: Report recruiting site(s) and dates, prospective versus retrospective collection, exact patient count per group, inclusion/exclusion and diagnostic criteria, and whether each participant contributed exactly one specimen.**
- **TODO: Report available demographics and clinical covariates, including age, sex, disease stage, treatment status, and comorbidities; summarize missingness and state which fields cannot be released.**
- **TODO: Report blood volume, collection tubes, time to processing, cell-isolation and cryopreservation procedures, quality-control thresholds, and sample/cell exclusion counts.**
- **TODO: Report assay/platform and reagent versions, sequencing instruments and depth, 293-probe panel design, and a verified protein-target manifest or explicitly state why only anonymized probe identifiers can be released.**
- **TODO: Report barcode-matching rates, APT normalization and filtering, all remaining software versions and parameters, and whether annotation was performed before investigators examined APT profiles or disease labels.**
- **The documented final cell-flow accounting is 374,854 linked post-QC cells – 13,062 Unknown cells = 361,792 benchmark cells. TODO: Data owners: provide participant and cell counts at all earlier collection/QC exclusions before a complete cohort-flow diagram can be drawn. Supply the approval, consent, access, and licensing evidence listed in the Ethics and Reproducibility Statements.**

B.2 BENCHMARK INTERFACE AND RELEASE CONTRACT

The cell-typing interface records pseudonymous cell and patient IDs, fixed folds, feature IDs, reference and predicted labels, and ontology columns. A complete release requires the processed paired matrices, permissions and access procedure, reference configurations, and a raw-data-to-evaluator command. The current split manifest, selected artifact checksums and rebuild commands cover only part of that release. **TODO: Finalize anonymous data/code access, licenses, original training environment locks, archival versioning, and the portable end-to-end command. Where access is controlled, provide the complete schema and access procedure.**

B.3 HIERARCHY AND CONDITIONAL-PAIRING PROTOCOL

Design and training conditions. This analysis is exploratory: its design followed inspection of a single-seed pilot. At both RNA widths, log-count normalization and mean-binned dispersion selection use only the current inner or outer training partition. All 361,792 cells are barcode-aligned. RNA is an internal reconstruction reference because it also supplies the operational labels.

Table 5 specifies both training recipes. For the matched probe, weights use only the current training subset. Writing N for its cell count, $n_{p(i)}$ for the count from the patient of cell i , and n_{y_i} for its class

count, the implemented weights are

$$a_i = \frac{1}{n_{p(i)} n_{y_i}}, \quad w_i = \frac{a_i}{N^{-1} \sum_{j \in \text{train}} a_j}.$$

This is the mean-normalized product of inverse patient frequency and balanced-class weights. The product does not ensure equal total weight per patient or per class. The full-budget MLP instead uses unweighted CE.

Setting	Matched SGD	Matched MLP	Full-budget MLP
Model / loss	One-vs-rest logistic; weighted log loss	64-unit ReLU; weighted cross-entropy	256/128-unit ReLU; dropout 0.1; unweighted cross-entropy
Training cells	Same fixed subset, at most 1,000 cells per training patient		All available training cells
Optimizer / rate	Averaged SGD; optimal schedule $\eta_t = 1/[\alpha(t + t_0)]$	Adam; initial rate 10^{-3}	AdamW; rate 10^{-3}
Update size / epochs	One cell per update; 40 passes; no tolerance stopping	Batch 512; 40 epochs; no early stopping	Batch 1,024; select epoch 1–50 on inner validation
Regularization grid	L2 $\alpha \in \{10^{-4}, 10^{-3}\}$	L2 $\alpha \in \{10^{-3}, 10^{-1}\}$	Weight decay $\{10^{-5}, 10^{-3}\}$
APT input	$\log(1 + \text{raw APT count})$, then featurewise training mean/SD standardization		Provided apt_expression, then training mean/SD standardization; no added log transform
RNA input	Per-cell counts scaled to 10^4 , log1p; training-selected 2,000/5,000 HVGs; training SD scaling without centering		Not used
Noise control	293 independent $\mathcal{N}(0, 1)$ features appended to scaled RNA; no further noise scaling; separate train/test draws		Not used

Table 5: Core experiment configurations read from the implemented recipes. The SGD offset t_0 follows the library’s optimal-schedule heuristic. Regularization parameters have estimator-specific meanings, not a shared penalty scale. Each outer fold uses the next frozen patient fold for inner validation; highest validation SB-Macro-F1 selects the grid value (and full-budget epoch), retaining the first candidate/earliest epoch on ties. Refits reinitialize on the outer development partition. Transform statistics, HVG selection and sample weights use only the current training partition; per-cell RNA library normalization does not estimate population parameters. The upstream transform underlying provided APT expression remains a provenance requirement, not an assumed raw-count equivalence.

Oracle and prior controls. Oracle decoding masks frozen fine scores to children of the true parent without changing routing or retaining only correctly predicted parents. Conditional majority and proportion priors use the actual training subset’s subtype frequencies within that parent. The proportion prior contributes its expected confusion matrix, which is then scored; this is not expected finite-sample F1. These controls are privileged-label diagnostics, not deployable predictions. For an individual lineage, restrict true cells to that lineage but retain all prediction columns, so predictions outside the lineage remain false negatives. Normalize each eligible patient’s within-lineage matrix, pool and score its fixed child classes. Patients with zero scope support are excluded. This differs from Table 1, where normalization uses each patient’s entire evaluated cell population.

Conditional shuffling. The targeted 2,000-HVG experiment uses three shuffle realizations per training seed for each control: within patient and within patient and true lineage. Entire APT vectors are permuted separately in training and test partitions, preserving each group’s empirical APT

distribution. The latter control uses privileged true test lineage. Singleton and unchanged-pair fractions are retained in the reproducibility outputs. Each realization has its own selected/refitted control model; it is not just a test-time perturbation of the paired model.

Statistical target and exact resampling order. Let $C_{s,m}^{(p)}$ denote the confusion matrix for training seed s , condition m and patient p in the chosen scope, with mass n_p . For patient multiplicities $w = (w_1, \dots, w_{40})$, define

$$F_{s,m}(w) = \text{MacroF1}_{\mathcal{K}} \left(\sum_{p:n_p>0} w_p C_{s,m}^{(p)} / n_p \right),$$

where $\mathcal{K}$ is the full ontology or the fixed child classes for a lineage. Dividing this pooled matrix by its total weight would not change F1. There is no pooling of confusion matrices across training seeds or shuffle realizations before computing F1. The paired-versus-shuffled point estimate is

$$\hat{\Delta} = \frac{1}{3} \sum_{s=1}^3 \left[F_{s,\text{paired}}(\mathbf{1}) - \frac{1}{3} \sum_{r=1}^3 F_{s,\text{shuffle}(r)}(\mathbf{1}) \right].$$

For each of 2,000 bootstrap replicates:

1. Draw 40 patients with replacement from the 40 held-out patients, giving one multinomial multiplicity vector w shared across conditions, seeds and shuffle realizations in the paired comparison.
2. Draw three seed indices s_1, s_2, s_3 with replacement from the three observed training seeds; a replicate does not select just one seed.
3. For each selected seed slot j , draw three shuffle indices with replacement from its three realizations. Compute each F1 separately on the shared patient draw, average these three control F1 values, and subtract that average from $F_{s_j,\text{paired}}(w)$.
4. Average the three seed-slot differences. Report the 2.5th and 97.5th percentiles of finite replicates. A scope with no eligible resampled patient is undefined, not assigned zero.

For non-shuffled comparisons the control has one value per seed, so step 3 does not resample a control realization. Full-budget MLP and RNA-width intervals likewise average seed-specific F1 values after paired patient resampling. Frozen-reference intervals in Table 15 resample patients only. These pointwise intervals do not correct for multiple comparisons, repeat model selection, or account for all analysis adaptation; three seeds provide limited training-uncertainty evidence.

RNA-width sensitivity. The 2,000/5,000-HVG reference grids use three seeds but one patient-shuffle realization per seed; their point estimates are not substitutes for the three-shuffle targeted contrast in Table 3. The 5,000-HVG paired-versus-RNA result is retained regardless of outcome. The more restrictive conditional shuffle was evaluated only at 2,000 HVGs; its unresolved overall interval was not followed by additional training to seek significance. Full run schedules and the pre-existing follow-up selection rule are documented in the reproducibility protocol.

Model	Scope	APT	APT oracle	RNA oracle	Prior MAP	Prior expected CM
SGD	All	0.089	0.289	0.542	0.148	0.182
SGD	B	0.133	0.334	0.556	0.176	0.199
SGD	Myeloid	0.147	0.201	0.661	0.072	0.124
SGD	NK	0.220	0.840	0.791	0.434	0.494
SGD	Other	0.018	0.470	0.603	0.496	0.487
SGD	T	0.119	0.184	0.379	0.067	0.098
MLP	All	0.093	0.309	0.672	0.148	0.182
MLP	B	0.212	0.405	0.750	0.176	0.199
MLP	Myeloid	0.162	0.211	0.774	0.072	0.124
MLP	NK	0.241	0.849	0.839	0.434	0.494
MLP	Other	0.076	0.497	0.652	0.496	0.487
MLP	T	0.119	0.191	0.535	0.067	0.098

Table 6: Matched-budget hierarchy diagnostics: mean over three training seeds, 2,000 training-selected RNA HVGs, identical capped training subsets and patient folds. All-cell rows normalize each patient’s complete matrix; lineage rows instead normalize within that lineage and score fixed children. Their scores cannot be averaged to recover all-cell SB-Macro-F1. Oracle and prior controls receive true lineage. Expected-CM F1 is not expected finite-sample F1. Other has 38 eligible patients; other scopes have 40. These are descriptive point estimates, not paired APT–RNA superiority tests.

HVGs	Task	Model	RNA	Paired	Noise	Patient shuffle	Paired minus RNA
2,000	Coarse	SGD	0.745	0.751	0.745	0.750	+0.0063 [0.0004, 0.0127]
2,000	Coarse	MLP	0.871	0.872	0.868	0.874	+0.0008 [-0.0057, 0.0075]
2,000	Fine	SGD	0.432	0.450	0.411	0.442	+0.0185 [0.0033, 0.0332]
2,000	Fine	MLP	0.589	0.607	0.573	0.597	+0.0180 [0.0092, 0.0298]
5,000	Coarse	SGD	0.772	0.777	0.773	0.777	+0.0047 [0.0016, 0.0079]
5,000	Coarse	MLP	0.873	0.881	0.876	0.878	+0.0086 [0.0036, 0.0144]
5,000	Fine	SGD	0.456	0.478	0.437	0.468	+0.0217 [0.0104, 0.0339]
5,000	Fine	MLP	0.596	0.612	0.586	0.606	+0.0165 [0.0093, 0.0289]

Table 7: RNA-width reference sensitivity: three training seeds and five patient folds. Point estimates average seed-specific SB-Macro-F1; paired differences have joint seed/patient 95% intervals. Patient shuffle uses one draw per seed in this grid; the targeted 2,000-HVG experiment uses three draws per seed for each shuffle type. All normalization and HVG selection are training-partition-only. Noise adds 293 dimensions. Differences are computed before rounding; differences and interval endpoints use four decimal places.

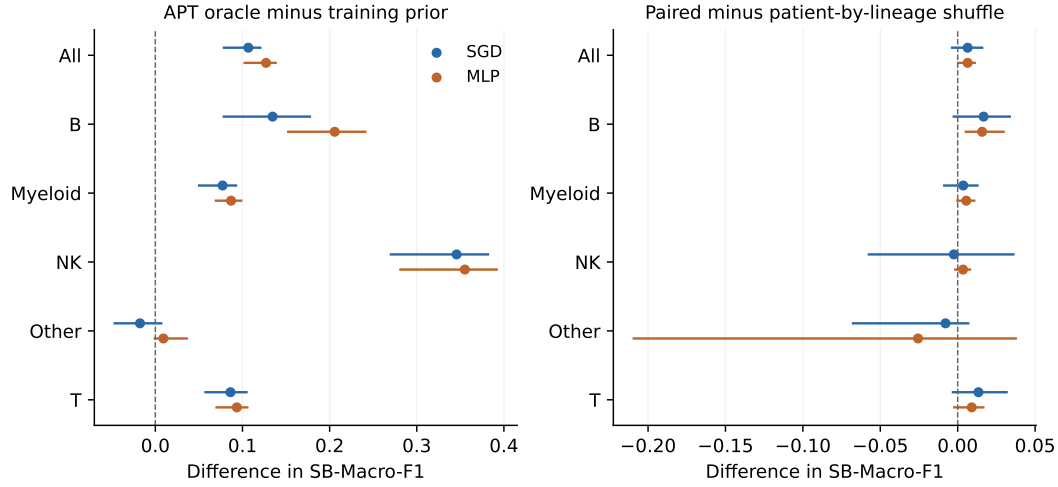


Figure 3: Exploratory scope-specific contrasts at the matched 2,000-HVG budget. Left: APT oracle minus the conditional-proportion expected-CM prior. Right: paired RNA+APT minus patient-by-lineage-shuffled APT. Pointwise intervals follow the joint resampling specified above; all lineages are retained. All-cell and within-lineage scores have different patient-normalization populations. Axes differ across panels. Overall right-panel intervals include zero; subgroup contrasts do not override that result.

B.4 FULL-BUDGET HIERARCHY SUPPORT

The full-budget 256/128-unit MLP uses all development cells and a different optimization recipe from the capped 64-unit matched-input MLP. Its overall ordinary/oracle/expected-prior scores are 0.141/0.385/0.180. The oracle-minus-prior difference is +0.205 ([0.174, 0.217]). Figure 4 reports every lineage. Differences from the matched-input experiment cannot be attributed to training-cell budget alone, because APT preprocessing, architecture, optimization and selection also differ (Table 5).

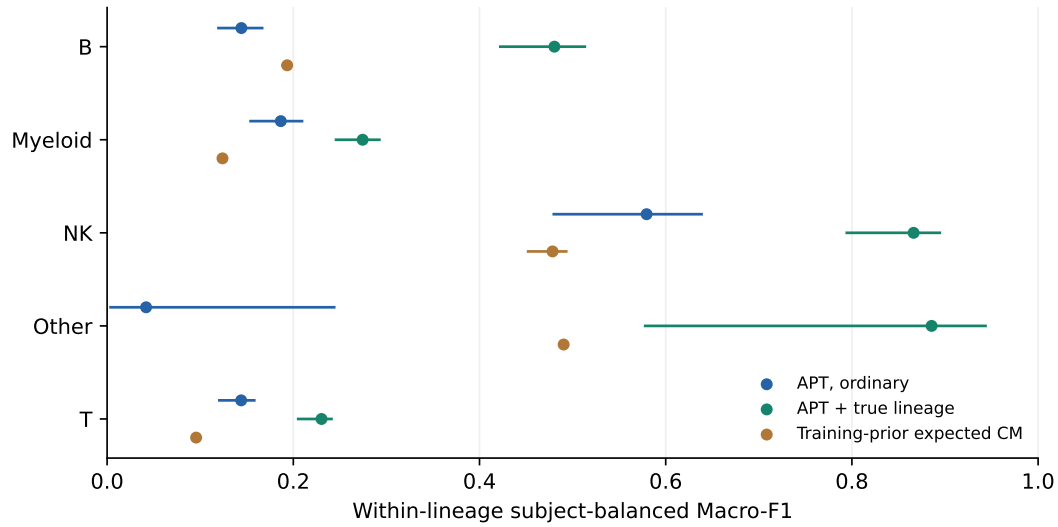


Figure 4: Full-budget MLP support: three-seed means and pointwise joint seed/patient 95% intervals. Each eligible patient’s matrix is normalized within the displayed lineage, then pooled before Macro-F1 is computed over its fixed children. Outside-lineage predictions remain false negatives. These scores cannot be averaged across lineages to reconstruct Table 15. The conditional-proportion baseline is F1 of an expected confusion matrix, not expected F1. Oracle information is privileged. This is not the capped-budget APT–RNA comparison in Figure 2.

B.5 TRUE-LINEAGE RESTRICTED DECODING

We replay the 15 frozen final checkpoints (three neural references, five patient-disjoint folds), retain explicit cell IDs and ontology columns, and require exact reconstruction of their original coarse and fine predictions before computing this diagnostic. For each of the same 361,792 held-out cells, we mask the fine score vector to subtypes whose parent is the true lineage, then take its argmax. We do not change Soft-cascade routing or condition the evaluation set on a correct coarse prediction. The diagnostic therefore uses privileged test-label information and cannot be placed on the deployable leaderboard. Table 8 reports subject-balanced pooled Macro-F1 and paired patient-bootstrap differences, conditional on these frozen models. Reduced candidate-set size contributes to the gain; neither the gain nor the remaining error identifies an assay-specific information ceiling.

Model	Ordinary fine	True-lineage oracle	Paired gain [95% CI]
Flat Transformer	0.139	0.366	+0.227 [0.202, 0.238]
HCE Transformer	0.143	0.361	+0.218 [0.196, 0.231]
Soft-cascade Transformer	0.145	0.358	+0.214 [0.194, 0.230]

Table 8: Privileged true-lineage restricted decoding on the same frozen predictions. Scores are subject-balanced pooled Macro-F1; gains use 2,000 paired patient-bootstrap resamples and pointwise intervals, conditional on fitted models. The oracle knows the true parent and is not a deployable method.

B.6 COMPLETE PER-SUBTYPE REPORTING

Table 9 retains all 27 subtypes, cell counts, patient coverage, and historical cell-weighted per-class F1 with 2,000 patient-bootstrap intervals. These are not the subject-balanced main metric. They support transparent class-level inspection, not a biological interpretation of marginal support-performance correlations.

Table 9: Complete subtype support and cell-weighted per-class OOF F1 (not SB-F1), with pointwise 95% patient-bootstrap intervals. Counts and patient coverage refer to the same 361,792 cells.

Subtype	Lineage	Cells	Patients	Soft-cascade	XGBoost
Erythrocyte	Other	65	6	0.253 [0.000, 0.395]	0.133 [0.000, 0.545]
Atypical Memory B	B	781	34	0.005 [0.000, 0.014]	0.005 [0.000, 0.014]
CD4 Monocyte-4	Myeloid	1,405	35	0.027 [0.006, 0.053]	0.014 [0.000, 0.033]
platelet	Other	1,574	38	0.078 [0.016, 0.125]	0.063 [0.015, 0.106]
Tcm	T	2,121	36	0.000 [0.000, 0.000]	0.004 [0.000, 0.016]
Memory B-2	B	2,130	33	0.017 [0.004, 0.033]	0.011 [0.000, 0.024]
Plasma	B	2,173	40	0.611 [0.522, 0.684]	0.599 [0.496, 0.685]
pDC	Myeloid	2,615	40	0.029 [0.010, 0.061]	0.019 [0.005, 0.040]
CD14 Monocyte-3	Myeloid	2,977	32	0.258 [0.170, 0.306]	0.177 [0.103, 0.211]
CD8 + CTL-3	T	3,244	27	0.020 [0.004, 0.035]	0.023 [0.008, 0.043]
GCB	B	3,283	31	0.039 [0.007, 0.071]	0.047 [0.001, 0.073]
Cycling T	T	3,667	40	0.028 [0.015, 0.043]	0.012 [0.003, 0.022]
CD14 Monocyte-2	Myeloid	3,931	40	0.075 [0.046, 0.102]	0.078 [0.056, 0.097]
CD8 + CTL-2	T	4,049	33	0.069 [0.017, 0.116]	0.078 [0.015, 0.113]
CD16 Monocyte-2	Myeloid	5,142	40	0.111 [0.064, 0.161]	0.117 [0.063, 0.164]
CD8 + CTL-1	T	7,303	40	0.009 [0.000, 0.024]	0.002 [0.001, 0.004]
cDC2	Myeloid	7,376	40	0.132 [0.099, 0.181]	0.125 [0.079, 0.167]
CD8 Tem-2	T	10,548	40	0.019 [0.005, 0.038]	0.017 [0.008, 0.030]
CD16 Monocyte-1	Myeloid	14,678	40	0.152 [0.111, 0.198]	0.118 [0.090, 0.150]
CD4 + Tcm-2	T	16,861	39	0.188 [0.111, 0.236]	0.188 [0.136, 0.220]
Naive T	T	17,960	40	0.131 [0.067, 0.183]	0.149 [0.073, 0.199]
Memory B-1	B	23,722	40	0.034 [0.020, 0.048]	0.032 [0.017, 0.047]
CD8 Tem-1	T	26,714	40	0.187 [0.128, 0.250]	0.162 [0.107, 0.221]
CD14 Monocyte-1	Myeloid	29,871	40	0.437 [0.378, 0.495]	0.451 [0.392, 0.507]
CD56 NK	NK	38,442	40	0.398 [0.320, 0.436]	0.437 [0.373, 0.480]
CD16 NK	NK	58,507	40	0.383 [0.326, 0.434]	0.376 [0.317, 0.429]
CD4 + Tcm-1	T	70,653	40	0.417 [0.368, 0.460]	0.435 [0.393, 0.473]

B.7 SCORING UNITS AND SUPPORTING SPLIT COMPARISON

Metric definitions. Let $C^{(p)} \in \mathbb{R}^{K \times K}$ be the confusion matrix for held-out subject p and $n_p = \sum_{ij} C_{ij}^{(p)}$. We distinguish

$$C^{\text{CW}} = \sum_p C^{(p)}, \quad F_1^{\text{CW}} = \text{MacroF1}(C^{\text{CW}}), \quad (11)$$

$$C^{\text{SB}} = \frac{1}{P} \sum_p \frac{C^{(p)}}{n_p}, \quad F_1^{\text{SB}} = \text{MacroF1}(C^{\text{SB}}), \quad (12)$$

$$F_1^{\text{within}} = \frac{1}{P} \sum_p \text{MacroF1}(C^{(p)}). \quad (13)$$

We call these *cell-weighted pooled Macro-F1*, *subject-balanced pooled Macro-F1*, and *mean within-subject Macro-F1*. Subject balancing normalizes each subject confusion matrix to unit mass before pooling; it is neither equal-cell subsampling nor averaging per-subject Macro-F1. Released files retain the legacy column name `patient_balanced_macro_f1` for F_1^{SB} , but no different patient-balanced statistic is used.

The numerical split comparison appears in this appendix (Table 10). Its historical scores are cell-weighted in both columns, rather than the subject-balanced leaderboard metric.

Exact subject-normalized pooling is primary. The older equal-cell Monte Carlo sensitivity remains archived, rather than occupying a separate table.

The historical patient-disjoint XGBoost run uses 32 training and eight test patients per fold, without a validation partition; Soft-cascade uses 24/8/8 training/validation/test patients. Both cell-split runs instead use 253,254/54,269/54,269 cells, with all 40 patients represented in each partition. An audit of the two saved cell-level split manifests verified pairwise-disjoint training, validation and test cell-ID sets. Patient-disjoint scores pool 361,792 OOF cells; cell-split scores use 54,269 cells, not a common test set across protocols. The historical Soft-cascade CSV exports used in this split audit, both patient-disjoint pooled OOF and cell-split test predictions, retain patient and label fields but not cell IDs. This limits independent barcode linkage for those exports, not the manifest-based partition check, and does not imply actual cell overlap. The separate frozen-checkpoint oracle replay in Appendix B.5 retains explicit cell IDs; XGBoost prediction IDs were checked against the historical manifests. The split-audit reproducibility record reports full counts, source hashes and historical training-record limitations. This is a descriptive comparison, not an estimate of a pure leakage effect.

Task	Model	Patient-disjoint	Cell split	Observed difference
Coarse lineage	XGBoost	0.322	0.372	+0.050
Coarse lineage	Soft-cascade Transformer	0.326	0.417	+0.090
Fine subtype	XGBoost	0.143	0.212	+0.069
Fine subtype	Soft-cascade Transformer	0.152	0.260	+0.108

Table 10: Historical descriptive protocol comparison. Both score columns are cell-weighted pooled Macro-F1 (CW), not the subject-balanced metric of Table 1. Observed difference is cell split minus patient-disjoint, computed from unrounded saved predictions before display rounding. Training, validation and test allocations were not matched; the patient-disjoint column pools five outer folds, whereas cell split evaluates one subset of cells from already-seen patients. Protocol-difference uncertainty was not estimated. These related model/resolution contrasts are not independent replications or causal estimates of leakage.

B.8 ADDITIONAL REFERENCE INVENTORY

Table 11 supplies the three additional classical references; Table 1 contains the unified v2 primary models. These additional historical recipes are not trained under that protocol.

Model	CI	Coarse SB-Macro-F1 $\uparrow$	Fine SB-Macro-F1 $\uparrow$
Linear SVM	P	0.309 [0.292, 0.324]	0.105 [0.089, 0.118]
Random Forest	P	0.253 [0.227, 0.277]	0.089 [0.079, 0.097]
Reference Correlation	P	0.171 [0.147, 0.192]	0.068 [0.050, 0.078]

Table 11: Additional historical APT-only references, not trained under the unified protocol of Table 1. P denotes pointwise 95% patient-bootstrap intervals conditional on frozen fits.

B.9 UNIFIED REFERENCE PROTOCOL AND COMPLETED FOLLOW-UP

The unified v2 comparison replaces the historical reference-recipe leaderboard in Table 1, not the separate prior-controlled and paired-RNA diagnostics. The protocol was fixed before generating the v2 outer predictions; the existing outer folds had already been examined, so this is an exploratory follow-up rather than an untouched confirmatory evaluation.

Common data and isolation. The input is the barcode-aligned raw APT count matrix with 293 columns in fixed source order. Each count is transformed by $\log_1 p$, then standardized feature-wise

using only the current fit partition. All annotated benchmark cells are used, without patient caps, class weights, patient weights or replacement sampling. Equal patient weight is used for scoring, not training. For test fold f , validation is $(f + 1) \bmod 5$ and the other three folds form the inner fit. A new scaler and a newly initialized model are fitted on all development patients after selection. Outer-test cells are never used for fitting the scaler, choosing a configuration or determining stopping. Table 12 gives exact observed counts.

Fold	Inner fit	Validation	Outer refit	Test
1	24 / 226,431	8 / 68,381	32 / 294,812	8 / 66,980
2	24 / 207,172	8 / 86,239	32 / 293,411	8 / 68,381
3	24 / 206,704	8 / 68,849	32 / 275,553	8 / 86,239
4	24 / 221,600	8 / 71,343	32 / 292,943	8 / 68,849
5	24 / 223,469	8 / 66,980	32 / 290,449	8 / 71,343

Table 12: Exact unified v2 patient / cell counts, reconstructed from input IDs and frozen partitions and checked against every recipe. All methods share these counts. Validation is the next cyclic outer fold; remaining patients form the inner fit. Outer refit combines inner fit and validation. Counts include all annotated benchmark cells, without a cap.

Model-specific search spaces. LR uses unweighted multinomial logistic regression with LBFGS, a maximum of 1,000 iterations and $C \in \{0.01, 0.1, 1, 10\}$. XGBoost uses CPU histogram trees (four requested threads), depth $\{3, 6\}$, learning rate $\{0.03, 0.1\}$, and tree counts $\{100, 300, 500\}$, with row and column subsampling both 0.8. All twelve combinations are fitted separately; this is not continuous tree-wise early stopping. LR and XGBoost independently select coarse and fine configurations by fixed-ontology validation SB-F1, then refit using the chosen C or tree configuration.

The plain MLP has 256/128-unit ReLU hidden layers, dropout 0.1 and separate linear fine/coarse heads. All Transformer variants share the existing APT feature-identity/value tokenizer and pre-norm encoder, with hidden dimension 128, two layers, four attention heads, feed-forward dimension 256 and dropout 0.1. These dimensions differ from the historical 256-dimensional, four-layer references in Appendix A. Neural models use AdamW with the full grid $\text{lr} \in \{3 \times 10^{-4}, 10^{-3}\}$ and weight decay $\in \{10^{-5}, 10^{-3}\}$. Batch size is 1,024 for MLP and 128 for Transformers. Each candidate trains for at most 50 epochs, with validation fine SB-F1 after every epoch and patience eight. Strict improvement updates the best epoch; ties retain the earliest epoch and first candidate. Fresh outer refits run for that selected epoch count without test monitoring. The v2 loop uses float32, no gradient clipping and no learning-rate schedule. The historical mixed-precision and contrastive warm-up settings are not used here.

Objectives and component scope. MLP, Flat and Cascade minimize unweighted fine CE plus 0.5 coarse CE. Cascade retains the global path, lineage experts and log compatibility term in Eq. 6, with temperature 1, expert scale 1, $\epsilon = 0.1$ and constant $\gamma = 0.5$ (no ramp). Flat bypasses experts and routing. The wrapper retains the common projection head and unused expert parameters for Flat; effective capacity and compute are therefore not identical. HCE instead minimizes the leaf and parent negative log subtree masses, with weights 1 and 0.5, over the joint 32-node softmax. Fine prediction selects among leaves; parent prediction sums each internal node and its descendants. No main-table model uses disease labels as training supervision; the inherited outer folds remain disease-stratified.

For the additional Flat/Cascade contrastive variants, normalize projected embeddings to unit length. With P_i the same-subtype, different-disease training examples in the mini-batch, the added objective is

$$0.1\mathcal{L}_{\text{con}}, \quad \mathcal{L}_{\text{con}} = -\frac{1}{|V|} \sum_{i \in V} \frac{1}{|P_i|} \sum_{j \in P_i} \log \frac{\exp(u_i^\top u_j / 0.1)}{\sum_{a \neq i} \exp(u_i^\top u_a / 0.1)}. \quad (14)$$

Here V contains anchors with positives; a mini-batch without valid anchors contributes zero. All non-self examples enter the denominator, including same-subtype/same-disease examples. This common v2 loss is distinct from the historical positive-mass objective in Appendix A.3. No disease labels are required at inference. Each of the four recipes selects its own validation optimum

under the same grid: the resulting contrasts compare whole training/selection procedures, not a routing-only intervention.

Seeds, scoring and intervals. Selection is performed once per model/fold (and per classical task) with seed 17; randomized final fits use seeds 17, 29 and 43. LR is deterministic and is fitted once per task/fold. A seed controls model initialization and, for neural models, a without-replacement cell permutation each epoch; it does not change the folds or number of unique training cells. Scores first pool patient-normalized confusion matrices over the complete OOF set, then average the resulting seed-specific F1 values, not predictions. Fine and coarse ontologies always contain 27 and five labels respectively, with zero-denominator F1 set to zero.

The bootstrap uses 5,000 replicates and RNG seed 20260915. Each replicate draws 40 patients with replacement and three seed indices with replacement. The same patient draw and matched seed slots are used across all conditions; F1 is computed separately in each seed slot and then averaged. LR reuses its one fit, so its uncertainty reflects only patient resampling. Paired differences and the 2×2 interaction are computed within each replicate, with percentile 95% intervals. These are pointwise, exploratory intervals, conditional on the chosen configurations, folds and finite seed set; they do not repeat model selection or correct for multiplicity. Shared development sets also prevent interpreting the OOF patients as independent fitted models. No equivalence margin or equivalence test is supplied, so overlapping or zero-containing intervals do not establish equivalence.

Deployable versus privileged decoding. Let p_k be the frozen fine probabilities, q_m the frozen parent probabilities and $g(k)$ the fixed parent mapping. D0 takes $\arg \max_k p_k$. D1 restricts that argmax to $g(k) = \arg \max_m q_m$. D2 takes the argmax of $q_{g(k)} p_k / \sum_{j: g(j)=g(k)} p_j$ (a 10^{-30} denominator floor). D4 restricts the argmax to the true reference parent. No weights change, and D4 is not used to select a model or report deployable performance. These rules use subtype probabilities and a fixed ontology; they are not new fitted hierarchical models. All conditions, including unfavorable decoding results, are retained in Table 13.

Model	Ordinary D0	Hard D1	Consistent D2	Oracle D4
Logistic Regression	0.134	0.110	0.116	0.360
XGBoost	0.134	0.126	0.131	0.362
Plain MLP	0.137	0.131	0.134	0.376
Flat Transformer	0.147	0.137	0.142	0.371
HCE Transformer	0.146	0.142	0.146	0.371
Soft-cascade (no disease)	0.148	0.136	0.143	0.371
Flat + contrastive	0.148	0.137	0.143	0.372
Cascade + contrastive	0.150	0.138	0.143	0.376

Table 13: Frozen-score decoding diagnostics for all unified v2 recipes (fine SB-F1, seed means). D1 restricts candidates to the predicted parent; D2 combines predicted parent probabilities with within-parent subtype probabilities. Both are deployable post-processing rules using the same frozen scores. D4 uses the true parent and is privileged, not a deployable score or an information upper bound. LR/XGBoost use separately selected parent models; neural references use their auxiliary parent outputs, with subtree probabilities for HCE. No retraining or test-based tuning of the decoding rules is performed. Full scores, seed values and paired intervals are released in the v2 CSV summaries.

Artifact validation and HCE readout correction. All 130 final score arrays are checked against the original cell IDs, patient IDs, labels, fixed class order and outer-fold membership without inter-section filtering. Recorded fit counts and input hashes agree across recipes; inner and outer standardization statistics are independently reconstructed from the raw inputs. The selected candidate and earliest best epoch are checked against saved validation histories. During aggregation, the initial HCE export was found to normalize internal-node logits alone for its parent readout. This does not match its subtree-supervised objective. Fifteen frozen HCE checkpoints were replayed to recover joint-node subtree probabilities, without retraining, reselection or changing fine predictions. Only coarse, D1 and D2 results are affected; D0 and D4 remain unchanged. The original exports, check-

point hashes and correction record are retained. This was a readout correction, not a new training experiment.

Compute and reproducibility artifacts. Runs were split between vllab11 and vllab15, both using RTX 6000 Ada GPUs for neural fits; LR/XGBoost remained on CPU. Per-fit audit records identify hardware, exact counts, seeds, selected parameters and search/refit timing. Table 14 summarizes realized budgets. The repository directory `analysis/unified.v2/` contains the frozen protocol, training and scoring scripts, source-validation script, HCE replay, machine-readable stage inventory, per-seed scores, all paired contrasts and source hashes. These supplement rather than resolve the existing data-access requirements. Historical scores and diagnostics remain separately identified in the following appendices; they are not pooled with v2 fits.

Recipe	Candidate fits	Refit epochs	Search h	Refit h
Logistic Regression	40	—	0.53	0.20
XGBoost	120	—	4.32	2.31
Plain MLP	20	15–26	0.08	0.04
Flat Transformer	20	14–45	4.73	3.79
HCE Transformer	20	14–37	4.10	3.34
Soft-cascade (no disease)	20	11–39	5.07	3.84
Flat + contrastive	20	11–47	5.85	6.19
Cascade + contrastive	20	19–46	6.23	7.03

Table 14: Realized v2 search/refit records summed across folds (and across tasks for LR/XGBoost). Neural search uses four candidates per fold; LR uses four and XGBoost twelve per task/fold. Epoch ranges are selected stopping epochs, not the epochs actually consumed by all search trials. Times are accumulated timed-block wall hours, not elapsed project time or GPU-hours: parallel jobs overlap, hosts differ, and input/scaler preparation is outside these blocks. Neural runs used RTX 6000 Ada GPUs; classical runs used CPU. This is a reproducibility record, not a cross-hardware efficiency ranking. HCE correction inference is excluded.

B.10 HISTORICAL REFERENCE TRAINING PROTOCOLS

Table 15 retains the earlier provided-expression scores that support the original MLP diagnostics. They are not v2 comparison rows.

Model	CI	Coarse SB-Macro-F1 $\uparrow$	Fine SB-Macro-F1 $\uparrow$
Logistic Regression	P	0.298 [0.281, 0.315]	0.126 [0.107, 0.139]
XGBoost	P	0.325 [0.302, 0.345]	0.137 [0.120, 0.149]
Plain MLP (3 seeds)	SP	0.334 [0.311, 0.355]	0.141 [0.125, 0.155]
Flat Transformer	P	0.326 [0.300, 0.348]	0.139 [0.114, 0.155]
HCE Transformer	P	0.335 [0.310, 0.356]	0.143 [0.121, 0.160]
Soft-cascade Transformer	P	0.332 [0.306, 0.354]	0.145 [0.122, 0.160]

Table 15: Historical provided-expression APT-only comparison (not unified v2): five patient-disjoint folds, 40 patients, 361,792 cells. All inputs are APT. Entries are subject-balanced pooled Macro-F1 with pointwise 95% intervals (2,000 resamples). CI type P conditions on frozen fits and resamples patients; SP resamples both training seeds and patients for the MLP three-seed mean. These uncertainty targets differ; interval widths are not directly comparable evidence of model stability. Comparisons are descriptive: training recipes and Transformer auxiliary supervision differ (Tables 16 and 4).

Table 16 separates the data used to select a recipe from the final model’s fit data. Counts come from keyed cell/patient records and saved splits, not equal-size-fold approximations. All six references evaluate the same 361,792 cells in five outer patient partitions with the fixed five-lineage/27-subtype ontology and the SB pooling in §3.3. Historical neural exports lack cell IDs, but the separate exact-prediction checkpoint replays retain them and were checked against the classical and MLP test sets without intersection filtering.

Model	Fit data	Validation / refit	Selection / training budget	Supervision / weights	Training seeds
Logistic Regression	D, all cells	None; one direct fit	Fixed retained recipe [†] : LBFGS, maximum 500 iterations	Separate coarse/fine CE; train-class balanced	No RNG used
XGBoost	D, all cells	None; one direct fit	Fixed retained recipe [†] : 400 trees, depth 6, learning rate 0.1	Separate coarse/fine labels; no sample weights	Not recovered
Plain MLP	Inner T; final D, all cells	V selects; fresh initialization and outer refit on D	Two decays (10^{-5} , 10^{-3}), 50 epochs each; task-specific SB-F1 selection	Separate coarse/fine CE; unweighted	20260912, 20260913, 20260914
Flat Transformer	T, all cells	V selects checkpoint; no D refit	Maximum 50 epochs, patience 10; fine CW-F1 on observed-label union	Coarse/fine CE; unweighted	42
HCE Transformer	T, all cells	V selects checkpoint; no D refit	Maximum 50 epochs, patience 10; fine CW-F1 on observed-label union	Two-level HCE; unweighted	42
Soft-cascade Transformer	T, all cells	V selects checkpoint; no D refit	Maximum 50 epochs, patience 10; fine CW-F1 on all 27 classes	Coarse/fine CE plus disease-based contrastive term; unweighted	42

Table 16: Historical training protocols for Table 15. D: 32 development patients, 275,553–294,812 cells; T: 24 inner/Transformer training patients, 206,704–226,431 cells; V: eight validation patients, 66,980–86,239 cells. Each outer test partition has eight patients and 66,980–86,239 cells. No training-cell cap is applied. [†]Complete original classical tuning/CLI records are unavailable: these settings are from surviving fit code, not a recovered search log. XGBoost’s source default seed is 42, not an independently verified original invocation. Neural seed values are recovered from stored configurations; bootstrap seeds are not training seeds.

The recipes use the provided 293-feature `apt_expression` matrix, not the raw-count-plus-`log1p` input of the matched modality probes. The available audit does not recover the complete upstream transformation that produced this supplied expression. Verified training-partition scaling therefore does not independently establish that every upstream operation was cell-local or fitted without test patients. This is a provenance limitation, not evidence that test information was actually used. Historical loaders fit feature-wise mean/SD on the fit partition only; MLP fits a new StandardScaler in each inner fit and outer refit. Saved MLP feature IDs and outer-refit scaler statistics were verified against the keyed input and development cells. The original classical estimator/scaler snapshots and complete historical search histories are unavailable; surviving code documents their fixed recipes, not an independently recovered tuning log. Historical neural feature order is supported by exact keyed replay, not a separately serialized original feature-ID manifest. No disease label enters the LR, XGBoost, MLP, Flat or HCE training objective.

The machine-readable audit provides exact fold/seed/stage counts, selected MLP settings, checkpoint provenance and unrecoverable fields. Earlier neural configurations sometimes permit five head-calibration epochs, but all 15 final parameter dictionaries match the validation-selected pre-calibration checkpoints exactly; replay disables calibration. Complete discarded-run compute and search histories are unavailable, so shared epoch limits do not establish equal total tuning cost. The P/SP intervals in Table 15 do not include repeated model selection.

B.11 FULL-BUDGET MLP FAILURE STRUCTURE

This analysis uses the same 15 saved fine-task score arrays (three seeds, five folds), all 361,792 OOF cells, and the full-budget APT preprocessing recipe of Table 16. Cell IDs, patient membership, true labels, score-column ontology, parent mapping and saved ordinary/oracle confusion matrices are checked without filtering to an intersection. No model is retrained or replayed. The existing expected-CM prior is independently reconstructed from outer-development subtype counts (SB-Macro-F1 0.180). The prior-controlled result in §5.2 and the full-budget support plot remain distinct from the capped RNA/APT probe.

Error categories and oracle transitions. For true subtype y , ordinary subtype prediction $\hat{y}$ and fixed parent map g , correct means $\hat{y} = y$, cross-parent error means $g(\hat{y}) \neq g(y)$, and within-parent error means $\hat{y} \neq y$ with $g(\hat{y}) = g(y)$. These exhaust all cells and do not use the independent coarse head. Within each patient and scope we divide counts by that scope’s cell count, then average eligible patients and seeds. The cross-error repair fraction instead divides corrected cross errors by cross errors within each patient, omitting patients with zero cross errors; all scope-eligible patients have cross errors in these runs. This is not a ratio of population-average rates. Other has 38 eligible patients; the other scopes have 40.

Oracle masks the same saved score vector to the true parent’s children, using the original column order and first-maximum tie rule. In all seeds, ordinary-correct cells remain correct and ordinary within-parent errors keep the same prediction. Cross-parent errors either become correct or remain wrong within the true parent. Patient-balanced ordinary correctness is 33.4%, and unconditional repaired mass is 30.8%, yielding 64.2% oracle correctness before rounding. These are correctness rates, not Macro-F1 values or its additive decomposition. Oracle is a privileged diagnostic, not deployable performance or an information upper bound.

Scope	Patients	Correct	Cross-parent	Within-parent	Oracle correct	Cross repaired
All	40	33.4	49.4	17.3	64.2	63.2
B	40	7.8	91.1	1.1	83.9	83.4
Myeloid	40	27.2	47.0	25.8	45.6	38.8
NK	40	39.3	57.0	3.8	90.9	89.2
Other	38	1.2	98.8	0.0	99.4	99.4
T	40	33.9	42.5	23.6	53.9	49.9

Table 17: Full-budget MLP error proportions (percent), averaged within patient and then across three seeds. The first three rates partition all cells in the scope and sum to 100% before rounding. Cross-parent uses the parent of the predicted subtype, not the independent coarse head. Cross repaired is the within-patient fraction of cross-parent errors corrected by oracle, averaged over patients with such errors (all listed patients in each seed); it is not a ratio of the displayed averages. These are error rates, not additive contributions to Macro-F1.

Residual confusion and support. We rank directed off-diagonal edges by their seed-mean subject-balanced confusion mass, with fixed ontology order breaking ties, and report the top three descriptively. CD8 Tem-1 $\rightarrow$ CD4 + Tcm-1 accounts for 6.64% of patient-balanced all-cell mass (seed range 6.50–6.82%); the reverse accounts for 2.76% (2.61–2.91%). They rank first/second in each seed. Their supports are 26,714 and 70,653 cells, respectively, both across all 40 patients. CD16 Monocyte-1 $\rightarrow$ CD14 Monocyte-1 ranks third overall (per-seed ranks 3/3/4), with 1.99% mass (1.88–2.08%); both appear in all patients (14,678 and 29,871 cells). These selected examples are not confined to rare classes, but abundance has not been causally controlled. Seed ranges are descriptive, not confidence intervals or subgroup significance tests.

Matrix and score conventions. For each seed we divide each patient’s full 27×27 confusion matrix by their total test-cell count and sum over patients (total matrix mass 40). Per-class F1 is computed on each such matrix and then averaged across seeds. The heatmaps instead average the three matrices before row normalization; their row fractions must not be used to reconstruct the mean-seed SB-F1. All classes and edges are retained, with zero rows marked rather than dropped (none occur here). Oracle’s zero cross-parent blocks are guaranteed by masking, not evidence that

the model learned lineage identity. Remaining confusions cannot distinguish annotation instability, measurement, preprocessing, training budget or model limitations without further evidence.

Subtype	Cells	Patients	Ordinary	Oracle	Largest oracle error (row %)
Atypical Memory B	781	34	0.001	0.347	Memory B-1 (48.1)
GCB	3,283	31	0.050	0.405	Memory B-1 (47.0)
Memory B-1	23,722	40	0.046	0.905	GCB (3.4)
Memory B-2	2,130	33	0.001	0.084	Memory B-1 (74.5)
Plasma	2,173	40	0.476	0.693	Memory B-1 (35.6)
CD14 Monocyte-1	29,871	40	0.458	0.610	CD16 Monocyte-1 (20.3)
CD14 Monocyte-2	3,931	40	0.107	0.211	CD14 Monocyte-1 (61.3)
CD14 Monocyte-3	2,977	32	0.248	0.341	CD14 Monocyte-1 (46.5)
CD16 Monocyte-1	14,678	40	0.148	0.425	CD14 Monocyte-1 (39.4)
CD16 Monocyte-2	5,142	40	0.090	0.193	CD14 Monocyte-1 (49.5)
CD4 Monocyte-4	1,405	35	0.006	0.008	CD14 Monocyte-1 (60.7)
cDC2	7,376	40	0.123	0.192	CD16 Monocyte-1 (40.4)
pDC	2,615	40	0.012	0.125	CD16 Monocyte-1 (46.4)
CD16 NK	58,507	40	0.393	0.939	CD56 NK (5.1)
CD56 NK	38,442	40	0.426	0.814	CD16 NK (21.3)
Erythrocyte	65	6	0.046	0.816	platelet (30.0)
platelet	1,574	38	0.051	0.993	Erythrocyte (0.0)
CD4 + Tcm-1	70,653	40	0.429	0.721	CD8 Tem-1 (11.9)
CD4 + Tcm-2	16,861	39	0.184	0.394	CD4 + Tcm-1 (18.9)
CD8 + CTL-1	7,303	40	0.000	0.036	CD4 + Tcm-2 (37.8)
CD8 + CTL-2	4,049	33	0.100	0.216	CD4 + Tcm-2 (34.8)
CD8 + CTL-3	3,244	27	0.013	0.026	CD4 + Tcm-2 (35.4)
CD8 Tem-1	26,714	40	0.221	0.344	CD4 + Tcm-1 (67.9)
CD8 Tem-2	10,548	40	0.008	0.054	CD4 + Tcm-1 (44.9)
Cycling T	3,667	40	0.022	0.192	CD4 + Tcm-1 (72.3)
Naive T	17,960	40	0.144	0.308	CD4 + Tcm-2 (29.3)
Tcm	2,121	36	0.005	0.010	CD4 + Tcm-1 (51.2)

Table 18: All 27 subtypes for the same full-budget MLP and three seeds. Ordinary/oracle entries average seed-specific per-class F1 derived from subject-normalized pooled confusion matrices. These are not the historical Soft-cascade CW values in Table 9. Destination percentages row-normalize the seed-mean oracle SB matrix; they are visual summaries, not per-patient conditional averages. Full matrices, per-seed F1 and every directed edge are retained.

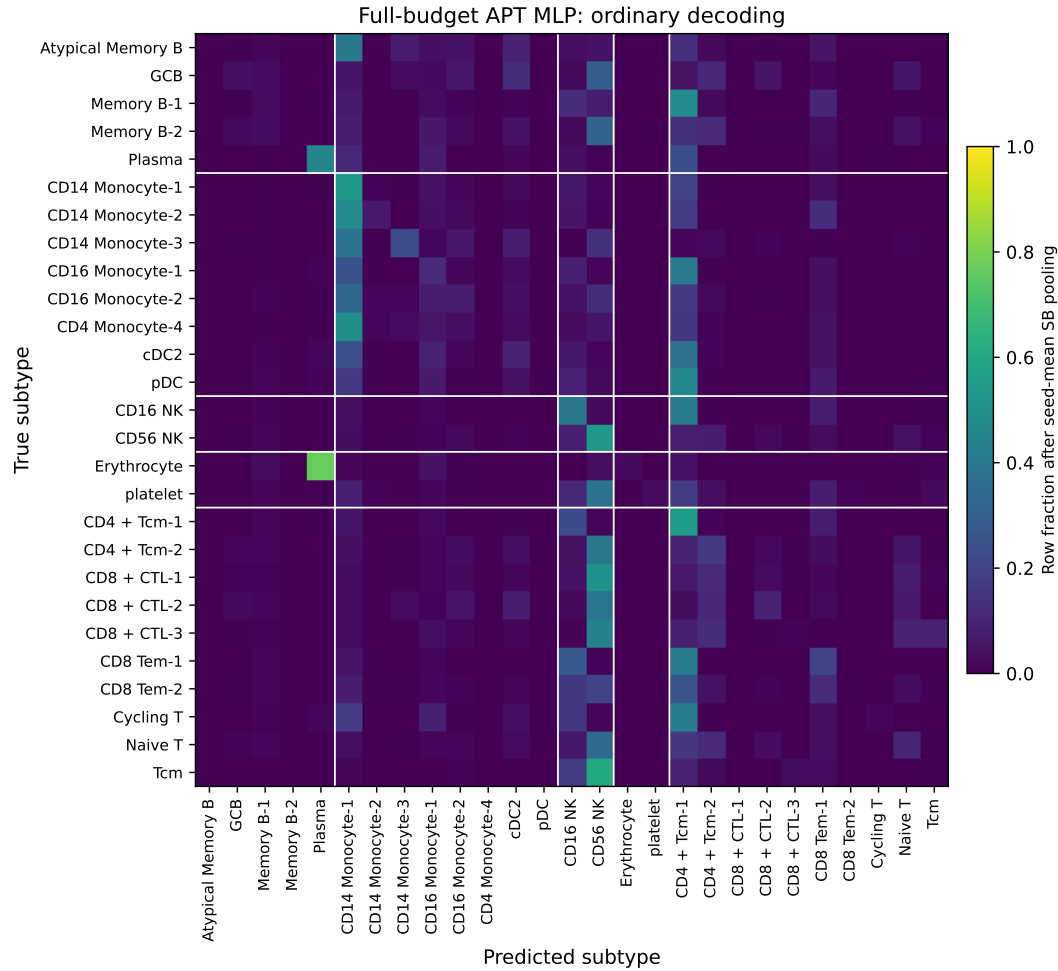


Figure 5: All 27 subtypes under ordinary full-budget MLP decoding. Rows are true and columns predicted subtype, ordered by lineage then fixed subtype order; white lines separate lineages. Color uses the same 0–1 scale as Figure 6. Each patient first has unit total mass, then seed matrices are averaged and rows normalized for visualization. This is not mean within-patient subtype recall and not a Macro-F1 matrix. All zero rows would be gray; none are present.

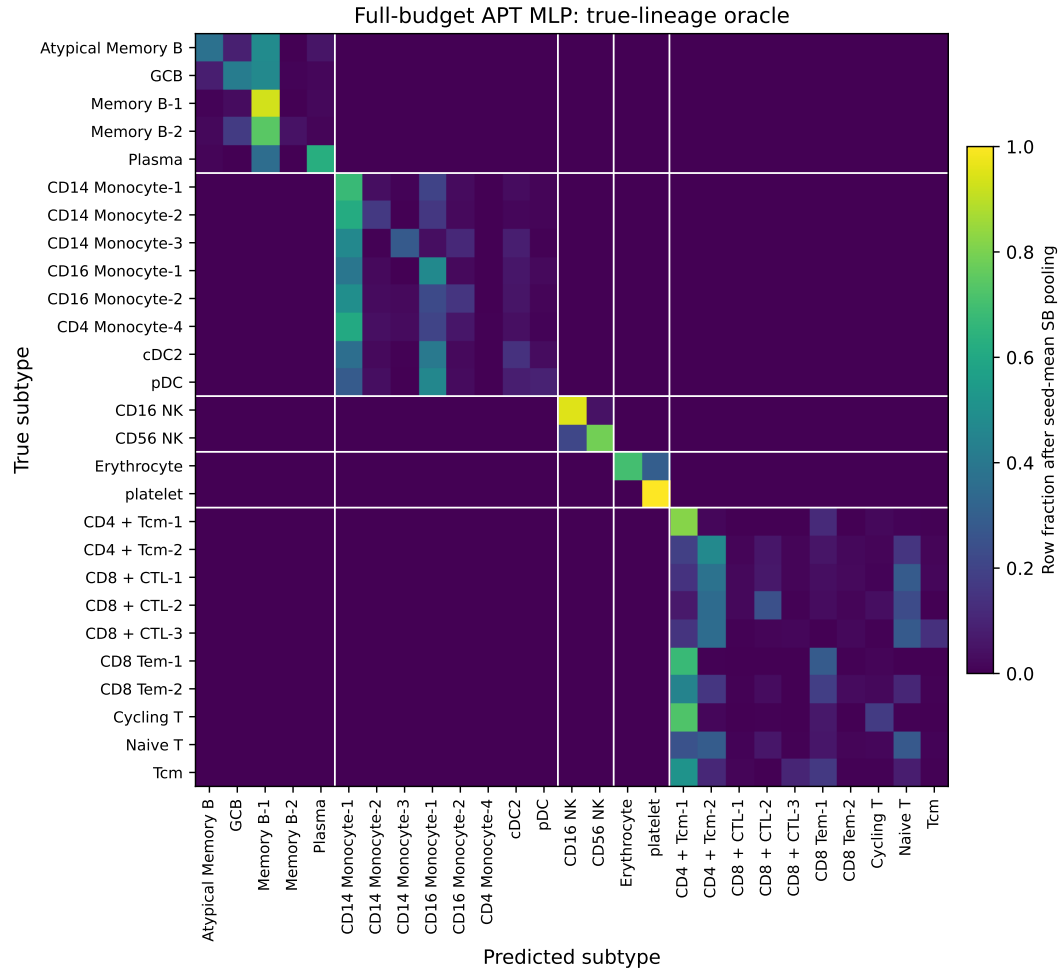


Figure 6: Full-budget MLP true-lineage oracle, using exactly the same scores, cells, three seeds, ordering and color scale as Figure 5. Cross-lineage blocks are zero by the decoding constraint, not a learned biological result. Within-lineage off-diagonal mass shows residual subtype confusion. The complete per-seed SB matrices, row-normalized matrices and directed edges are released with the analysis; heatmaps do not replace seed-specific scoring.

B.12 ARCHIVED ANALYSES

The version archive retains the earlier resource-comparison table, single-seed paired-input pilot, chance adjustment, disease-classification audits, subject–cell allocation and assay-budget analyses, marginal support correlations, and redundant cell-weighted metrics. Composition recovery (including direct ridge and the unresolved lineage result), the incomplete LODO stress test, and RNA-difficulty/rescue/damage analyses are archived in full rather than selectively reported. These analyses do not support the two core cell-identity findings reported here. The revision index identifies the source versions and retained outputs.